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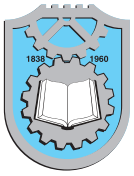
Forecasting Time-Varying Intermarket Dependencies Between Cryptocurrencies and Conventional Assets Using Machine Learning

Мастер рад

Крагујевац, 2026.

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Forecasting Time-Varying Intermarket Dependencies Between Cryptocurrencies and Conventional Assets Using Machine Learning

Мастер рад

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У оквиру овог мастер рада кандидат треба да истражи могућност прогнозирања временски-променљивих међутржишних зависности између криптовалута и конвенционалних финансијских актива применом модела машинског учења. У склопу тога кандидат треба да:

1. прикупи историјске дневне цене за Bitcoin и традиционалне активе (акцијске индексе, фондове племенитих метала, индекс америчког долара, Ethereum) и изврши одговарајућу предобраду и синхронизацију скупа података;
2. развије репродуцибилан рачунарски поступак за израчунавање временски-променљивих корелационих циљева помоћу клизних прозора различитих дужина и конструкцију предиктивних особина заснованих на инерцији зависности, волатилности и повратима;
3. реализује ванузорачну евалуацију и поређење неколико модела машинског учења (AR, ElasticNet, Ridge, Random Forest, GBM, XGBoost) са DCC-GARCH(1,1) економетријским репером применом процедуре проклизавајућег прозора и Diebold-Mariano теста статистичке значајности;
4. уведе практични сигнални слој за рано детектовање стресних дана на тржишту традиционалних актива на основу динамике криптовалута и прогноза зависности, и вреднује га метрикама класификационе тачности прикладним за неуравнотежен скуп класа.

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_____

Abstract

This thesis investigates whether information contained in cryptocurrency dynamics can be used to forecast time-varying dependency structures with conventional assets and, in a second step, to derive an early warning signal for investors. Bitcoin is treated as the base cryptocurrency, while the conventional asset universe includes equity indices, precious-metal exchange-traded funds, the U.S. dollar index, and Ethereum as an additional crypto reference. A reproducible machine-learning pipeline is developed for data acquisition, return construction, rolling-correlation target generation, feature engineering, walk-forward model evaluation, and benchmarking against a leakage-safe DCC-GARCH specification. The empirical results show that intermarket dependency is forecastable, but that most predictive power originates from persistence in the dependency series itself. Consequently, simple autoregressive baselines outperform more complex models on average, while machine-learning models still provide useful improvements relative to the DCC-GARCH benchmark. An investor-oriented signal layer based on crypto features and dependency forecasts achieves moderate but meaningful performance and is best interpreted as a risk-management overlay rather than a standalone market-timing engine.

Keywords: cryptocurrencies, Bitcoin, intermarket dependency, correlation, machine learning, DCC-GARCH, forecasting, risk management.

Резиме

У овом мастер раду разматра се проблем прогнозирања променљивих међутржишних зависности између криптовалута и традиционалних финансијских актива. У фокусу је Bitcoin као основна криптовалута, док скуп традиционалних актива обухвата акцијске индексе, фондове везане за племените метале и индекс америчког долара. Развијен је репродуцибилан рачунарски поступак који покрива преузимање података, израчунавање логаритамских приноса, формирање временски променљивих корелационих циљева, конструисање особина, ванузорачну процену више модела машинског учења и поређење са DCC-GARCH економетријским репером. Поред основног задатка прогнозирања зависности, уведен је и практични слој за формирање раног сигнала ризика који треба да упозори инвеститора на стресне дане на тржишту традиционалних актива. Резултати показују да је временски променљива међутржишна зависност предвидива, али да највећи део предвидивости потиче из сопствене инерције циљне серије, док сложенији модели машинског учења доносе ограничено додатно побољшање у односу на једноставне ауторегресионе приступе. Практични сигнални слој даје умерене резултате и има смисла пре свега као додатни филтар ризика, а не као самосталан механизам за свакодневно одређивање смера тржишта.

Кључне речи: криптовалуте, Bitcoin, међутржишне зависности, корелација, машинско учење, DCC-GARCH, управљање ризиком.

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List of Abbreviations

Abbreviation	Full form
AR1	First-order autoregressive model
AUC	Area under the ROC curve
BTC	Bitcoin (`BTC-USD`)
CSV	Comma-separated values
DCC	Dynamic Conditional Correlation
DCC-GARCH	Dynamic Conditional Correlation GARCH(1,1)
DM	Diebold–Mariano (test)
EDA	Exploratory Data Analysis
ElasticNet	Elastic Net regularized regression
ETF	Exchange-Traded Fund
ETH	Ethereum (`ETH-USD`)
F1	F1 score (harmonic mean of precision and recall)
GARCH	Generalized Autoregressive Conditional Heteroskedasticity
GBM	Gradient Boosting Machine
GLD	SPDR Gold Shares ETF (`GLD`)
GPU	Graphics Processing Unit
GSPC	S&P 500 index (`^GSPC`)
HAR	Heterogeneous Autoregressive (model)
IXIC	NASDAQ Composite index (`^IXIC`)
MAE	Mean Absolute Error
ML	Machine Learning
NASDAQ	National Association of Securities Dealers Automated Quotations
OOS	Out-of-sample
RF	Random Forest
RMSE	Root Mean Squared Error
ROC	Receiver Operating Characteristic
S&P 500	Standard & Poor's 500 index
SLV	iShares Silver Trust ETF (`SLV`)
UUP	Invesco DB US Dollar Index Bullish Fund (`UUP`)
XGB	XGBoost (Extreme Gradient Boosting)

1 Introduction

1.1 Motivation and research context

The progressive integration of cryptocurrency markets into the global financial system has elevated them from a technological experiment to a substantive channel of cross-market risk transmission. Bitcoin, in particular, has become a continuously traded, high-visibility asset whose price dynamics attract attention not only from crypto-native participants but also from institutional portfolio managers, macro strategists, and regulatory observers. As the market infrastructure matures and institutional participation deepens, a consequential research question emerges: does information embedded in cryptocurrency price dynamics carry predictive content for future conditions in conventional asset classes such as equities, metals, or currency-related instruments?

This thesis addresses that question by investigating *time-varying intermarket dependency*. Rather than attempting to forecast price levels or returns directly, the analysis models the evolving relationship between Bitcoin and conventional assets through rolling correlation structures and examines whether those structures are forecastable in a rigorous out-of-sample framework. The distinction is methodologically important. Cross-market correlation is well documented to be non-stationary, exhibiting pronounced temporal variation in periods characterized by liquidity stress, inflationary pressure, monetary tightening, or abrupt transitions in investor risk appetite. Under such conditions, the sign and magnitude of intermarket linkages may change at rates that conventional long-horizon static diversification assumptions fail to accommodate.

The practical motivation is equally compelling. A risk-oriented portfolio manager does not require a precise one-day-ahead return forecast to derive actionable intelligence. In many risk-management contexts, the more important information concerns early recognition that the market environment is undergoing a structural shift—that co-movement patterns are changing and that the conditional probability of a stress episode in conventional asset markets is rising. If cryptocurrency-derived information contributes measurably to such early-warning capacity, then even a modest and statistically bounded forecasting advantage constitutes a meaningful input to quantitative risk management.

1.2 Research questions

The central research problem is decomposed into two complementary layers. The first layer is scientific and methodological: are time-varying intermarket dependencies between Bitcoin and conventional assets forecastable in a disciplined out-of-sample framework, and how do machine-learning models perform relative to classical econometric benchmarks under rigorous walk-forward evaluation? The second layer is applied: can dependency forecasts derived from cryptocurrency dynamics be converted into a market-relevant warning signal capable of providing timely identification of elevated downside risk in conventional markets?

More specifically, the thesis addresses the following research questions:

1. Is the rolling dependency between Bitcoin and conventional assets forecastable out of sample?
2. Do machine-learning models outperform a leakage-safe DCC-GARCH benchmark?

3. Do machine-learning models add value beyond simple persistence-based baselines such as AR1?

4. Can crypto-derived dependency forecasts be converted into a practically useful stress-warning signal for conventional markets?

1.3 Contributions of the thesis

To address these questions, the thesis constructs a reproducible, modular forecasting pipeline implemented in Python. The pipeline ingests price data, computes log returns, constructs rolling-correlation targets across multiple window lengths, engineers lagged and volatility-based predictors, and evaluates a range of models within a walk-forward experimental design. The model set encompasses simple persistence benchmarks, regularized linear methods, tree-based ensemble methods, XGBoost, and a leakage-safe implementation of the DCC-GARCH benchmark. A second, investor-oriented layer is superimposed on the dependency-forecasting stage; this layer classifies stress regimes for conventional assets on the basis of crypto-derived features and forecasted dependency dynamics.

The empirical findings support a substantive but carefully bounded conclusion. Out-of-sample dependency forecasting is feasible; the target is not unpredictable noise. However, the dominant source of predictability is the strong serial persistence inherent in the dependency process itself, and consequently the most competitive baseline models are autoregressive and naive persistence variants. This result is not a deficiency of the framework; it is a substantive empirical finding. It establishes that machine-learning value in this context is incremental and comparative rather than transformative, and that any claim of dramatic forecasting gains must be evaluated against appropriately strong persistence baselines. The second-layer signal model demonstrates moderate but statistically meaningful predictive value, functioning most effectively as an auxiliary risk filter rather than as a comprehensive directional predictor.

The contribution of the thesis is therefore threefold. First, it constructs a clean, reproducible framework for forecasting rolling intermarket dependency between cryptocurrency and conventional asset markets. Second, it provides a structured, methodologically rigorous comparison between classical econometric modeling and a representative set of machine-learning approaches under walk-forward evaluation. Third, it extends the dependency-forecasting analysis to an applied investor-warning layer, thereby establishing an explicit and transparent bridge between the statistical forecasting problem and a practically relevant decision-making context.

1.4 Thesis outline

The remainder of the thesis is organized as follows. Section 2 reviews the theoretical foundations of time-varying dependence modeling, financial contagion, machine-learning forecasting methods, and the statistical framework for model comparison. Section 3 presents the dataset, target construction methodology, feature engineering procedure, model specifications, and the evaluation framework. Section 4 reports the empirical results for both the dependency-forecasting layer and the investor signal layer. Section 5 interprets the findings in depth, examines what the results do and do not imply, and acknowledges the principal limitations of the study. Section 6 concludes and proposes directions for future research.

2 Theoretical Background and Literature Review

2.1 Why intermarket dependency matters

Intermarket dependency occupies a central position in modern financial theory because the benefits of diversification, the propagation of contagion, the efficacy of hedging strategies, and the measurement of portfolio risk all depend fundamentally on the co-movement structure among assets. In the canonical mean-variance framework, dependence is summarized by a constant correlation coefficient. Empirical evidence, however, has long established that this static representation is inadequate. Pairwise correlations are time-varying, respond asymmetrically to positive and negative shocks, and tend to rise sharply during market downturns—precisely the episodes in which diversification is most urgently required [1]. Consequently, the forecasting of *time-varying dependence* is both theoretically motivated and practically indispensable for risk management.

The relevance of dynamic dependence is amplified when one of the assets under study is cryptocurrency-based. Bitcoin trades on a continuous, globally accessible market, responds rapidly to shifts in sentiment and funding liquidity, and is simultaneously subject to a broad range of economic narratives: inflation-hedging demand, speculative risk appetite, technological optimism, regulatory uncertainty, and global macro positioning. This multiplicity of drivers means that Bitcoin occupies no single, stable role in a multi-asset portfolio. At different points in the business and credit cycle, it may exhibit the characteristics of a speculative high-beta growth instrument, a liquidity-sensitive risk proxy, or an idiosyncratic sentiment barometer. That structural instability renders its relationship with conventional assets uniquely suitable for a time-varying analytical approach.

2.2 Cryptocurrencies and conventional assets

The academic literature on the intersection of cryptocurrency markets and conventional asset classes has expanded substantially over the past decade, developing along several distinct research strands. One strand examines Bitcoin as a standalone asset and characterizes its return distribution, volatility clustering, tail risk profile, and degree of weak-form informational efficiency. A second strand investigates portfolio diversification and asks whether Bitcoin provides economically meaningful diversification benefits relative to equity indices, gold, crude oil, or major fiat currencies. A third strand, most directly relevant to the present thesis, studies connectedness, contagion, and return and volatility spillovers across markets, with particular attention to crisis and stress episodes.

A recurring finding across these strands is structural instability bouri2017hedge, corbet2018exploring. The dependence between Bitcoin and broad equity indices tends to be statistically weak or directionally inconsistent during tranquil market conditions but strengthens materially during episodes of pronounced risk-on or risk-off positioning ji2018network, kristoufek2015main. Precious metals exhibit generally lower and more variable co-movement with Bitcoin dyhrberg2016bitcoin, klein2018bitcoin, a pattern attributable to their distinct investor bases and established safe-haven role. Currency-related instruments introduce a further dimension, as they serve as proxies for global funding liquidity corbet2018exploring.

These empirical regularities motivate the modeling strategy adopted in the present thesis. Rather than imposing a static dependence coefficient or estimating a single full-sample

correlation, the thesis models dependence as a time-varying process. This specification accommodates gradual regime drift, abrupt structural breaks, and heterogeneity in the pace and direction of co-movement changes across window lengths and asset pairs.

2.3 Rolling correlation as a forecasting target

The rolling window correlation estimator is conceptually transparent and computationally tractable. For two return series $\{r_{1,t}\}$ and $\{r_{2,t}\}$, the rolling correlation at time t over a window of length w is defined as

$$\hat{\rho}_t^{(w)} = \frac{\sum_{s=t-w+1}^{t}(r_{1,s} - \bar{r}_{1,t})(r_{2,s} - \bar{r}_{2,t})}{\sqrt{\sum_{s=t-w+1}^{t}(r_{1,s} - \bar{r}_{1,t})^2} \cdot \sqrt{\sum_{s=t-w+1}^{t}(r_{2,s} - \bar{r}_{2,t})^2}}, \tag{1}$$

where $\bar{r}_{k,t}$ denotes the in-window mean of series k. Traversing this estimator across all t yields a time series $\{\hat{\rho}_t^{(w)}\}$ that characterizes the temporal evolution of intermarket co-movement.

The rolling correlation target possesses several properties that make it attractive in a forecasting study. It is directly interpretable: positive values signal co-directional movement, while negative values indicate a hedge-like relationship. It is comparable across asset pairs and window lengths, enabling a systematic multi-asset analysis. It focuses attention on changes in market structure rather than on the level dynamics of individual return series.

The primary methodological challenge imposed by this target is strong serial persistence. Because $\hat{\rho}_t^{(w)}$ is constructed from overlapping observation windows, successive estimates share $w - 1$ common observations, inducing high autocorrelation in the target series. This persistence is likely to dominate predictive performance and may prevent complex machine-learning models from contributing materially beyond simple autoregressive baselines. The present thesis treats this property not as an obstacle but as an analytically important empirical characteristic of the dependency process.

To improve the statistical behavior of the target, the thesis applies the Fisher z-transformation:

$$z_t^{(w)} = \frac{1}{2} \ln\left(\frac{1 + \hat{\rho}_t^{(w)}}{1 - \hat{\rho}_t^{(w)}}\right), \tag{2}$$

which maps the bounded correlation values $\hat{\rho} \in (-1, 1)$ to an approximately unbounded real-valued quantity whose sampling distribution is closer to Gaussian. This transformation is standard practice in the analysis of correlation-like quantities and stabilizes variance near the boundaries of the unit interval.

An important alternative to Pearson rolling correlation is the copula-based dependence framework patton2006copula, which separately models marginal distributions and the dependence structure, thereby capturing tail dependence and nonlinear co-movement that Pearson correlation cannot detect. The present thesis adopts rolling Pearson correlation for four reasons: (i) direct interpretability for risk-management practitioners; (ii) computational tractability within the walk-forward estimation framework; (iii) compatibility with the DCC-GARCH benchmark, whose output is the conditional correlation matrix R_t derived from the conditional covariance structure engle2001theoretical; and (iv) the Fisher z-transformation stabilizes the sampling distribution near the boundary of the

unit interval. The restriction to Pearson dependence is acknowledged as a limitation, and copula-based extensions are identified as a priority for future research.

2.4 Econometric benchmark: DCC-GARCH

The Dynamic Conditional Correlation GARCH model, introduced by Engle [2], provides the classical econometric benchmark for the present study. The DCC-GARCH framework decomposes the conditional covariance matrix H_t of a vector of returns $\mathbf{r}_t$ according to

$$H_t = D_t R_t D_t, \tag{3}$$

where $D_t = \mathrm{diag}(h_{1,t}^{1/2}, h_{2,t}^{1/2})$ is a diagonal matrix of conditional standard deviations and R_t is the time-varying conditional correlation matrix. Each conditional variance $h_{k,t}$ follows a univariate GARCH(1,1) process,

$$h_{k,t} = \omega_k + \alpha_k \varepsilon_{k,t-1}^2 + \beta_k h_{k,t-1}, \tag{4}$$

where $\varepsilon_{k,t} = r_{k,t}/h_{k,t}^{1/2}$ is the standardized residual. The standardized residuals $\tilde{\varepsilon}_t = D_t^{-1} r_t$ are then used to update the pseudo-correlation matrix Q_t via

$$Q_t = (1 - a - b)\bar{Q} + a\, \tilde{\varepsilon}_{t-1}\tilde{\varepsilon}_{t-1}^{\top} + b\, Q_{t-1}, \tag{5}$$

where $\bar{Q}$ is the unconditional covariance of standardized residuals, and the scalar parameters $a \geq 0$, $b \geq 0$, $a + b < 1$ govern the speed of mean reversion and the persistence of conditional correlation. The conditional correlation matrix is recovered as

$$R_t = \mathrm{diag}(Q_t)^{-1/2}\, Q_t\, \mathrm{diag}(Q_t)^{-1/2}. \tag{6}$$

DCC-GARCH is conceptually well suited to this thesis because it explicitly models time-varying co-movement under heteroskedastic conditions, which are characteristic of both cryptocurrency and conventional equity returns. However, the model's reliance on a specific parametric structure may limit its adaptability when the data exhibit nonlinear interactions or structural breaks. Furthermore, implementations that estimate the model on the full sample prior to computing out-of-sample predictions inadvertently exploit future information. To eliminate this source of leakage, the present thesis employs a strictly walk-forward re-estimation protocol in which the DCC-GARCH model is re-fitted on an expanding window at each evaluation step.

2.5 Machine-learning approaches to forecasting

Machine learning is relevant to financial dependency forecasting not because it provides a guaranteed performance advantage, but because it offers a class of models capable of capturing nonlinear patterns, threshold dynamics, and multivariate interaction effects that are difficult to specify parametrically. In this thesis, machine-learning methods are evaluated against both the DCC-GARCH econometric benchmark and simple persistence baselines, with the dual objective of characterizing their relative strengths and identifying the conditions under which they contribute incremental predictive value.

2.5.1 The HAR model and persistence-based forecasting

The Heterogeneous Autoregressive (HAR) model, introduced by corsi2009har, provides a natural benchmark for forecasting persistent financial time series. The HAR model approximates long-memory dynamics by summing three autoregressive components operating at daily, weekly, and monthly horizons:

$$z_t = \phi_0 + \phi_d z_{t-1} + \phi_w \bar{z}_{t-5:t-1} + \phi_m \bar{z}_{t-22:t-1} + \varepsilon_t, \tag{7}$$

where $\bar{z}_{t-k:t-1}$ denotes the arithmetic mean of z over the preceding k periods. This specification is particularly well-suited to rolling correlation targets, which inherit a moving-average smoothing structure from the overlapping observation windows used in their construction andersen2003realized. The absence of HAR from the present model set constitutes a recognized limitation; its inclusion is reserved for future work, where a systematic comparison of HAR against AR1 and machine-learning methods across multiple forecast horizons would be informative.

2.5.2 Regularized linear models

Regularized linear models, including Ridge regression and Elastic Net, represent strong baselines in structured tabular forecasting settings. Ridge regression imposes an ℓ_2 penalty on the coefficient vector, shrinking estimates toward zero to reduce estimation variance without introducing sparsity. Elastic Net combines ℓ_1 and ℓ_2 regularization:

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \left\{ \frac{1}{n} \sum_{i=1}^{n} (y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2 + \lambda \left[(1-\alpha) \|\boldsymbol{\beta}\|_2^2 + \alpha \|\boldsymbol{\beta}\|_1 \right] \right\}, \tag{8}$$

where $\lambda > 0$ is the overall regularization strength and $\alpha \in [0, 1]$ governs the relative contribution of the two penalties [3]. The ℓ_1 component promotes sparsity and implicit variable selection, which is desirable when many lagged predictors carry weak or redundant signal. These models are particularly well suited to the present feature set, which contains multiple transformations of highly correlated return and volatility series.

Despite their linear functional form, regularized models frequently perform competitively in smooth, persistence-driven forecasting problems. When the signal-to-noise ratio is modest and the dominant feature is a lagged value of the target itself, the bias introduced by linearity is often outweighed by the variance reduction achieved through regularization.

2.5.3 Tree-based ensemble models

Random Forests construct an ensemble of decision trees, each trained on a bootstrapped subsample and a random subset of features, and aggregate their predictions to reduce variance and improve generalization [4]. Gradient Boosting Machines instead build trees sequentially, with each new tree fitted to the pseudo-residuals of the current ensemble [5]. XGBoost extends this boosting framework with second-order gradient approximations, ℓ_1 and ℓ_2 regularization terms on tree structure, and computational optimizations that facilitate large-scale application [6].

Tree-based ensembles are natural candidates for the present forecasting problem because the relationship between cryptocurrency-derived features and dynamic intermarket dependence may involve nonlinearities and interaction effects. The marginal impact of Bitcoin

realized volatility on future equity dependence, for instance, may differ substantially depending on whether the current correlation regime is already elevated. Tree ensembles can detect and exploit such conditional patterns without requiring the analyst to specify the interaction structure in advance.

That said, tree-based models are not uniformly superior in one-step-ahead forecasting of a highly persistent target. When the target is overwhelmingly governed by its most recent lagged value, more flexible models may contribute negligible additional signal while incurring a higher estimation variance penalty. This is a primary motivation for the disciplined walk-forward comparison presented in the thesis.

2.6 Walk-forward evaluation and information leakage

The temporal ordering of financial time series imposes strict requirements on evaluation design. Random train-test splits are inadmissible in this context because they violate the causal ordering of observations and introduce look-ahead bias, allowing a model trained on data from period $t+k$ to predict outcomes from period t. The only defensible evaluation strategy for time-series forecasting is one that ensures every prediction is generated using only information available at or before the forecast origin.

The thesis therefore employs a walk-forward procedure in which models are trained on an expanding historical window, a one-step-ahead prediction is produced, and the window is then extended by one observation. To balance computational cost against realism, model refitting is performed at periodic intervals rather than at every single step. This design mirrors the operational constraints of a real forecasting deployment and eliminates the possibility of inadvertent leakage.

The evaluation protocol is accorded the same level of importance as model selection. A complex model evaluated with implicit leakage may appear to substantially outperform a simpler, correctly evaluated competitor, producing a misleading benchmark comparison. Because the thesis aims to deliver defensible empirical conclusions, methodological integrity in evaluation is treated as a non-negotiable requirement.

2.7 Model comparison and the Diebold–Mariano test

Ranking models by average prediction error across the out-of-sample evaluation period provides useful descriptive information but does not establish whether observed performance differences are statistically distinguishable from sampling variation. The Diebold–Mariano test addresses this gap by providing a formal test of the null hypothesis of equal predictive accuracy between two competing forecasts [7].

Let $e_{1,t}$ and $e_{2,t}$ denote the forecast errors of models 1 and 2 at time t, and let $d_t = g(e_{1,t}) - g(e_{2,t})$ denote the loss differential under a loss function $g(\cdot)$ (typically squared or absolute error). The DM test statistic is

$$\mathrm{DM} = \frac{\bar{d}}{\sqrt{\hat{V}(\bar{d})\,/\,T}} \xrightarrow{d} \mathcal{N}(0,1), \tag{9}$$

where $\bar{d} = T^{-1}\sum_{t=1}^{T} d_t$ is the mean loss differential, $\hat{V}(\bar{d})$ is a heteroskedasticity- and autocorrelation-consistent (HAC) variance estimator, and T is the number of evaluation

periods. Under the null of equal predictive accuracy, the statistic is asymptotically standard normal.

In the context of this thesis, the DM test serves two distinct inferential purposes. First, it determines whether the best-performing machine-learning configuration yields a statistically significant improvement over the leakage-safe DCC-GARCH benchmark. Second, it establishes whether any apparent machine-learning advantage over simple autoregressive or naive persistence baselines survives formal statistical scrutiny. This second comparison is critical: a failure to significantly outperform an AR(1) baseline would imply that the bulk of forecastability derives from the serial persistence of the target series rather than from cross-asset feature learning.

2.8 From dependency forecasting to investor signaling

A study confined to the statistical forecasting of rolling correlation would constitute a self-contained academic contribution. However, the practical motivation of the present thesis demands an additional analytical layer. For the forecasting results to be actionable from an investor perspective, the dependency forecasts must be translated into a decision-relevant object that addresses the risk-management problem directly.

The thesis therefore introduces a second-stage signal layer. Rather than forecasting the next-day return direction of a conventional asset, the signal layer determines the probability that the next trading day will belong to a *stress regime*, defined by conditions of elevated downside risk in the conventional asset of interest. This formulation is more directly aligned with portfolio risk management, where the primary objective is not to predict every daily fluctuation but to provide timely advance warning of deteriorating market conditions.

The separation of forecasting and signaling into distinct stages also reflects a broader epistemological point. A model that is scientifically informative—in the sense of reducing forecasting error relative to a benchmark—need not be directly implementable as a trading rule, and conversely, a model engineered for binary classification of market stress may sacrifice statistical precision for decision-theoretic relevance. By making this separation explicit, the thesis establishes a transparent and intellectually honest bridge between the academic forecasting problem and the applied portfolio risk-management context.

2.9 Positioning of the present thesis

The present thesis occupies the intersection of financial econometrics, machine-learning methodology, and applied quantitative risk analysis. It does not advance the claim that cryptocurrency data can resolve the problem of market timing. Rather, it examines a more precisely formulated and empirically defensible claim: that cryptocurrency price dynamics contain statistically significant information about the evolution of intermarket dependency structures, and that this information can be used to construct a modest but economically meaningful early-warning signal for conventional asset markets.

This positioning is consequential for the interpretation of results. The primary scientific contribution of the thesis lies in establishing the forecastability of dynamic intermarket dependence and in providing a rigorous comparison of persistence baselines, machine-learning models, and the DCC-GARCH econometric benchmark under a consistent walk-

forward evaluation protocol. The investor signal layer extends this contribution into a practical direction while remaining anchored to what the empirical evidence can legitimately support.

3 Methodology

3.1 Research design

The thesis adopts a layered research design. The first layer is a regression-style forecasting problem in which the target is the next value of a transformed rolling correlation series between Bitcoin and another asset. The second layer is a classification-style problem in which the output of the first layer, together with crypto-derived features, is used to estimate the probability of a stress day for the conventional asset. This structure mirrors the distinction between scientific measurement and practical action.

The design is intentionally conservative. Rather than optimizing purely for in-sample fit, all important modeling decisions are embedded in a walk-forward evaluation framework. This includes target formation, feature alignment, benchmark estimation, and signal generation. The objective is to preserve temporal realism and minimize the risk of accidental look-ahead bias.

3.2 Asset universe and data source

The empirical analysis uses daily market data obtained through the `yfinance` interface [8]. Bitcoin, represented by `BTC-USD`, serves as the base cryptocurrency throughout the study. The comparison set includes the following assets:

Table 1: Asset universe used in the thesis.

Ticker	Interpretation
`BTC-USD`	Bitcoin, base cryptocurrency
`ETH-USD`	Ethereum, crypto reference asset
`^GSPC`	S&P 500 index
`^IXIC`	NASDAQ Composite index
`GLD`	Gold ETF
`SLV`	Silver ETF
`UUP`	U.S. Dollar Index ETF proxy

This universe was chosen for substantive rather than purely technical reasons. The two equity indices represent broad U.S. risk assets. Gold and silver ETFs capture precious-metal exposure, often linked to inflation concerns, safe-haven narratives, or commodity sentiment. The dollar index proxy reflects global dollar strength and defensive macro positioning. Ethereum is included as an additional crypto benchmark in order to contrast cross-crypto dependency with crypto-to-conventional dependency.

The sample spans from 9 November 2017 to 27 February 2026, yielding 2,976 daily observations per asset. The effective start date is determined by the availability of Ethereum price data on Yahoo Finance; earlier dates contain no ETH-USD observations and are therefore excluded from the aligned panel. The resulting dataset is balanced: all seven tickers share an identical date index with no missing observations after alignment and forward-filling of non-trading-day gaps up to two consecutive calendar days.

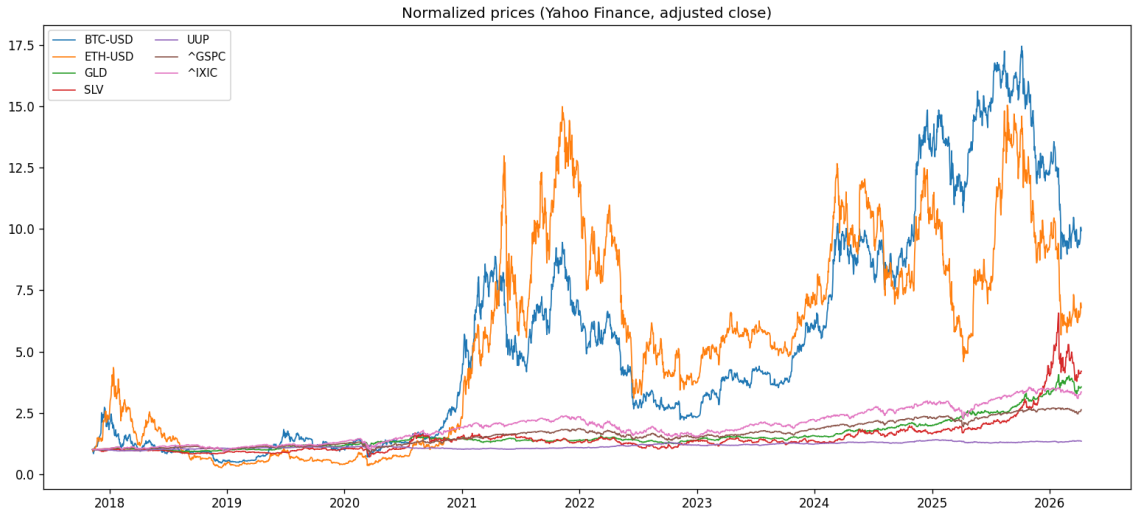


Figure 1: Normalized price evolution of all seven assets over the full sample period (2017–2026). Series are scaled to a common base of 1 at the start date to allow visual comparison of relative performance.

3.3 Return construction and preprocessing

All price series are converted into daily log returns. Log returns are preferred because they are additive over time and are standard in both econometric modeling and machine-learning pipelines for financial time series. For a price series P_t, the return is computed as

$$r_t = \log P_t - \log P_{t-1}.$$

Missing observations are handled through alignment across available dates, and all downstream targets and features are built only after the return matrix is synchronized.

A cache-aware preprocessing workflow is used in the codebase. If raw price data already exist, they are reused to guarantee reproducibility. If the structure of prices or the ticker universe changes, the processed returns are recomputed. This avoids stale intermediary files while keeping the workflow efficient.

3.4 Target construction: rolling dependency

For each pair consisting of Bitcoin and another asset, a rolling correlation series is computed over windows of 14, 30, 60, and 90 trading days. Let $r_t^{(b)}$ denote the return of Bitcoin and $r_t^{(a)}$ the return of the other asset. For a given window length w, the dependency target at time t is

$$\rho_t^{(w)} = \mathrm{corr}(r_{t-w+1:t}^{(b)}, r_{t-w+1:t}^{(a)}).$$

The resulting series is then transformed using the Fisher transformation

$$z_t^{(w)} = \frac{1}{2} \log \left(\frac{1 + \rho_t^{(w)}}{1 - \rho_t^{(w)}} \right) = \operatorname{arctanh}\left(\rho_t^{(w)} \right),$$

which maps the bounded correlation domain $(-1, 1)$ into an unbounded real-valued space. In the empirical implementation, the one-step-ahead value of the transformed dependency

series is used as the main forecasting target. The implementation clips the raw correlation away from the boundary to prevent numerical overflow:

Listing 1: Fisher z-transform and target construction (`pipeline.py`).

```
def fisher_z(r: pd.Series, eps: float = 1e-6) -> pd.Series:
    # Clip away from +/-1 to prevent arctanh overflow
    return np.arctanh(r.clip(-1 + eps, 1 - eps))

def build_target(returns, base, other, window, horizon):
    corr = returns[base].rolling(window).corr(returns[other]).dropna()
    # Shift by horizon to get the one-step-ahead target
    target = corr.shift(-horizon).dropna()
    return fisher_z(target)
```

A key implication of this design is that the target is already a smooth object. It is therefore much more persistent than a raw return series. This persistence is one of the defining empirical features of the thesis and is directly reflected in the later results.

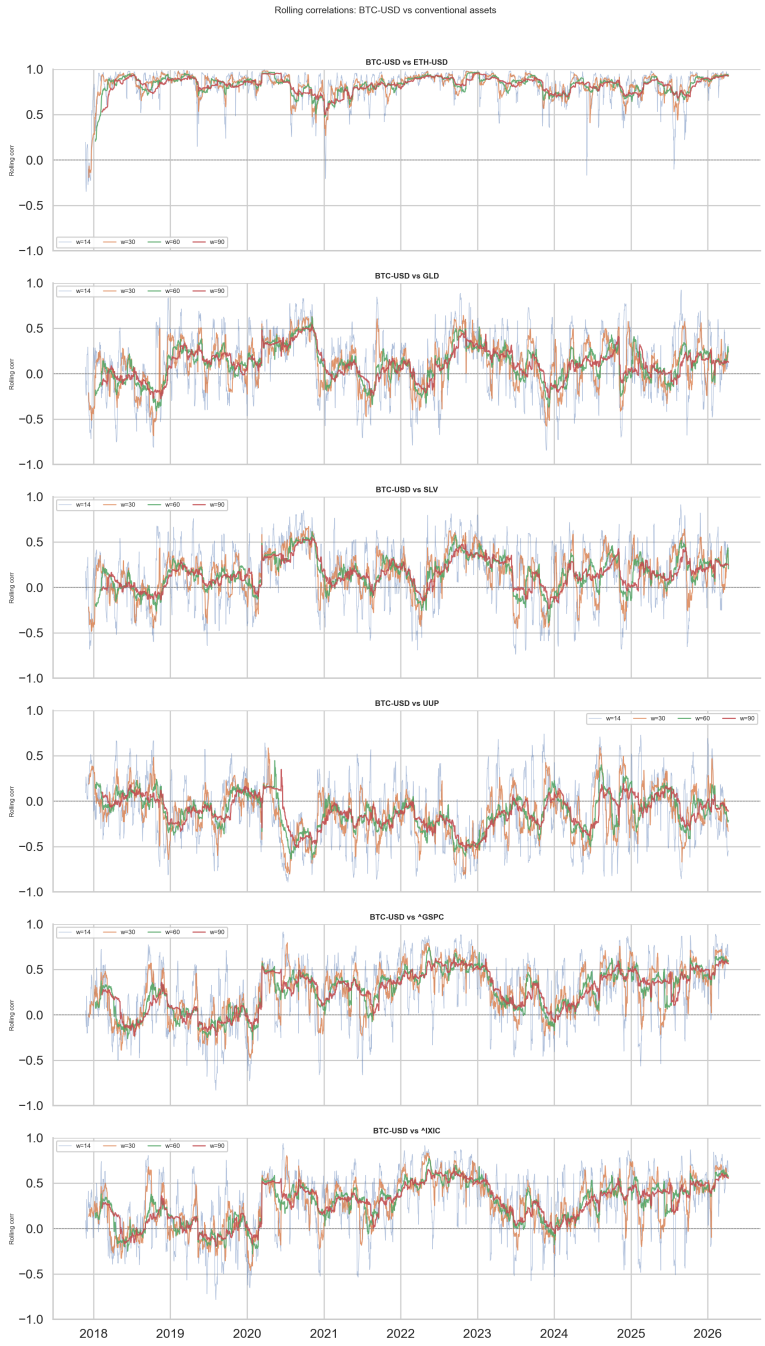


Figure 2: Rolling dependency between Bitcoin and each of the five conventional assets across all four window lengths (14, 30, 60, 90 days). The Fisher-transformed series illustrates the strong time-variation and persistence that motivate the forecasting framework.

3.5 Feature engineering

The feature set is intentionally structured around interpretable financial quantities. The goal is not to create an excessively large black-box predictor matrix, but to engineer a compact set of variables that can plausibly affect future dependency dynamics.

The features fall into several groups:

Table 2: Main groups of predictors used in the dependency-forecasting layer.

Group	Description
Dependency lags	Recent values of the rolling dependency target
Return lags	Lagged returns of Bitcoin and the paired asset
Rolling means	Local average return over short windows
Rolling volatility	Local standard deviation of returns
Spread features	Absolute and signed gaps between crypto and asset behavior
Correlation regime gap	Short-window minus long-window correlation

Dependency lags are especially important because the target is persistent by construction. Return lags allow the models to react to recent directional information. Rolling means and volatility measures summarize local regime conditions. The correlation regime gap (`corr_diff`) captures whether the short-term dependency is diverging from the longer-term baseline, which may signal an impending regime shift. The implementation of the feature matrix is shown in Listing 2:

Listing 2: Core feature engineering routine (`pipeline.py`).

```
# Dependency lags: the single most important predictor group
for lag in [1, 2, 5, 10]:
    features[f"dep_lag{lag}"] = y_current.shift(lag)

# Rolling volatility of both assets
features["vol_base"]  = returns[base].rolling(window).std().reindex(idx)
features["vol_other"] = returns[other].rolling(window).std().reindex(idx)

# Return lags for both assets
for lag in [1, 2, 5]:
    features[f"r_base_lag{lag}"]  = returns[base].shift(lag).reindex(idx)
    features[f"r_other_lag{lag}"] = returns[other].shift(lag).reindex(idx)

# Correlation regime gap: short-window minus long-window correlation
corr_short = rolling_corr(returns[base], returns[other],
                          max(5, window // 4)).reindex(idx)
corr_long  = rolling_corr(returns[base], returns[other],
                          window).reindex(idx)
features["corr_diff"] = corr_short - corr_long
```

The investor signal layer inherits the best dependency forecast available for the pair and combines it with crypto-derived predictors to form a smaller classification-oriented feature set. In practice, this means that the second layer is informed by both the current crypto state and the estimated direction of the dependency process.

3.6 Forecasting models

3.6.1 Baseline models

Two simple baselines are included in every forecasting experiment. The first is `Naive_Last`, which predicts the next dependency value using the most recent observed

dependency. The second is `AR1`, which estimates a one-lag autoregressive structure. These baselines are essential because the target is persistent. Any more complex model must therefore justify itself relative to these very strong simple alternatives.

3.6.2 Regularized linear models

The linear machine-learning block includes Ridge and Elastic Net. Standardization is applied before fitting these models, since coefficient regularization depends on feature scaling. Ridge is expected to perform well when many predictors are weak but correlated. Elastic Net is more aggressive and can be advantageous if only a subset of variables contains stable information.

3.6.3 Tree ensembles

The non-linear block includes Random Forest, GBM, and XGBoost. These models are fit in a walk-forward setting with periodic refits. XGBoost attempts GPU execution when available, but automatically falls back to CPU mode if the environment does not support the requested configuration. This makes the pipeline more robust and prevents a hardware-specific failure from invalidating the experiment.

The complete model set is initialized as follows:

Listing 3: Model definitions used in the walk-forward experiment (`pipeline.py`).

```
models = {
    "Naive_Last": None,            # persistence baseline (no fit required)
    "AR1":        LinearRegression(),
    "Ridge":      make_pipeline(StandardScaler(), Ridge(alpha=1.0)),
    "ElasticNet": make_pipeline(
                      StandardScaler(),
                      ElasticNet(alpha=0.01, l1_ratio=0.5, max_iter=5000)
                  ),
    "RF":  RandomForestRegressor(
               n_estimators=300, max_depth=8, n_jobs=-1
           ),
    "GBM": GradientBoostingRegressor(
               n_estimators=300, max_depth=4, learning_rate=0.05
           ),
    "XGB": xgb.XGBRegressor(
               device="cuda", n_estimators=300,
               max_depth=5, learning_rate=0.05,
               subsample=0.8, colsample_bytree=0.8
           ),
}
```

3.6.4 DCC-GARCH benchmark

A leakage-safe DCC-GARCH walk-forward benchmark is implemented as a separate module in the project. This is a key methodological correction relative to many simplified comparisons found in exploratory work. Instead of fitting DCC over the full sample and then evaluating it as if it were out of sample, the benchmark is repeatedly re-estimated

on the training window only, and forecasts are produced forward from that point. This makes the benchmark slower, but methodologically consistent.

3.7 Investor signal layer

The signal layer reframes the problem as an early warning classification task. The target is not generic next-day direction, but a *stress day* indicator for the conventional asset. A stress day is defined relative to the distribution of negative returns, using a configurable sigma-based threshold. This target design is intentionally conservative and aligns better with the practical goal of downside protection.

Three classifiers are evaluated in this layer: logistic regression, random-forest classification, and gradient-boosted classification. The probability output of each model is compared with a threshold to decide whether the investor should treat the next day as elevated-risk. This makes the output interpretable as a probability-guided warning rather than a binary black-box trade instruction.

3.8 Walk-forward estimation framework

All experiments use an expanding-window walk-forward design. Let the initial estimation sample contain the first T_0 observations. For each forecast origin $t \geq T_0$, the model is fit using observations up to time t, then used to produce a prediction for $t+1$. The process continues until the end of the sample. Periodic refitting is used to reduce computational burden while preserving causal ordering.

Listing 4: Expanding-window walk-forward loop (`pipeline.py`).

```
for t in range(min_train, n_obs):
    train_idx = np.arange(0, t)

    # Refit all models on the expanding training window periodically
    if (t - last_refit) >= refit_every:
        last_refit = t
        for name, model in model_specs.items():
            if model is not None:
                model.fit(X_values[train_idx], y_values[train_idx])
                fitted[name] = model

    # One-step-ahead prediction - uses only information up to t
    for name in fitted:
        preds[name][t] = fitted[name].predict(X_values[[t]])[0]
```

This design has several benefits. It mirrors real forecasting conditions, supports fair benchmark comparisons, and generates a full time series of out-of-sample prediction errors. The latter is useful not only for average metrics, but also for rolling error diagnostics and Diebold–Mariano testing.

3.9 Evaluation metrics

For the dependency-forecasting layer, performance is evaluated using mean absolute error (MAE), root mean squared error (RMSE), and the out-of-sample coefficient of determi-

nation R^2. RMSE is emphasized in model ranking because it penalizes large deviations more strongly and is well suited to smooth continuous targets.

For the investor signal layer, the primary metrics are Balanced Accuracy, F1 score for the downside class, area under the ROC curve (AUC), and exit rate. Balanced Accuracy is especially important because stress-day classification is an imbalanced problem. F1 for the downside class focuses attention on the quality of negative-event detection. Exit rate measures how often the signal would remove the investor from the market, thereby connecting prediction quality with practical usage frequency.

3.10 Model comparison methodology

Average metrics alone can be misleading when model differences are small. The thesis therefore applies the Diebold–Mariano test to compare forecast loss series between selected model pairs. The main comparison of interest is between the strongest non-benchmark machine-learning model and the DCC-GARCH benchmark. A secondary comparison contrasts that machine-learning model with the naive persistence baseline.

This setup is intentionally transparent. The thesis does not hide the fact that the best overall model may be a simple baseline. Instead, the DM framework is used to distinguish between three levels of claim:

1. whether dependency is forecastable at all,
2. whether machine learning improves upon a classical econometric benchmark,
3. whether machine learning adds value beyond persistence.

The empirical conclusions of the thesis are organized around precisely these three levels.

3.11 Implementation and reproducibility

The entire workflow is implemented in Python and organized as a modular codebase. The main pipeline coordinates loading configuration files, computing returns, generating targets, building features, fitting models, producing prediction files, saving figures, and exporting LaTeX-ready tables. Dedicated modules are used for DCC walk-forward benchmarking, notebook support, and the investor signal layer. This modular structure is important because the thesis is not merely a conceptual study but a reproducible computational artifact.

Outputs are stored in a structured directory hierarchy. Metrics, DM tests, and signal results are saved as CSV files; thesis-ready tables are exported to LaTeX; and visual diagnostics are saved as figures. This design reduces manual post-processing and helps align the research workflow with the writing workflow.

3.12 Methodological expectations

Before turning to the empirical results, it is useful to clarify what would count as success. Success in this thesis does not mean discovering a model that perfectly predicts all conventional markets from Bitcoin. That would be an unrealistic standard for any serious financial forecasting study. A more appropriate standard is the following:

- the dependency target should display out-of-sample forecastability,
- the benchmark comparison should reveal whether machine learning improves on DCC-GARCH,
- the investor signal layer should provide non-random and practically interpretable stress warnings.

These criteria are modest but scientifically defensible. They allow the thesis to make claims that are both empirically supported and intellectually honest.

4 Empirical Results

4.1 Overview of the empirical output

The pipeline generates a comprehensive, internally consistent set of empirical artifacts: out-of-sample performance metrics, model rankings, Diebold–Mariano test results, investor-signal classification statistics, prediction series, and diagnostic figures. The breadth of this output is deliberate. The empirical contribution of the thesis cannot be reduced to a single table; it rests on the coherence and cross-consistency of evidence examined from multiple complementary angles — average forecast accuracy, statistical significance of performance differentials, temporal error dynamics, and practical stress-warning quality.

Two overarching findings emerge from the aggregate evidence. First, the dependency process is forecastable in a rigorous out-of-sample walk-forward framework. Second, the primary source of forecastability is the strong serial persistence of the rolling correlation target rather than the extraction of novel cross-asset information. These two findings are complementary: together, they precisely characterize what the forecasting problem is and what it is not.

4.2 Average dependency-forecasting performance

Table 3 reports the average out-of-sample performance of the main models aggregated across all dependency experiments. Rankings are determined primarily by RMSE, the metric most sensitive to large deviations in a smooth, bounded target.

Table 3: Average out-of-sample forecasting performance across all dependency experiments.

Model	Average RMSE	Average MAE	Average R^2
AR1	0.0665	0.0405	0.9423
Naive_Last	0.0672	0.0395	0.9411
Ridge	0.0914	0.0636	0.8917
ElasticNet	0.0921	0.0643	0.8902
RF	0.1009	0.0709	0.8694
GBM	0.1035	0.0732	0.8630
XGB_GPU	0.1043	0.0744	0.8605
DCC_GARCH	0.2147	0.1687	0.3814

Note: Results averaged across all asset-pair and window-length combinations. RMSE and R^2 computed on out-of-sample walk-forward predictions. Best-performing model excluding DCC-GARCH is AR1.

Four notable patterns emerge from Table 3. First, the top-ranked model is the AR1 baseline, with an average RMSE of 0.0665 and an out-of-sample R^2 of 0.9423. Second, the naive persistence model is nearly indistinguishable from AR1 (RMSE difference of 0.0007), confirming that the dominant forecasting mechanism is autoregressive in nature. Third, the regularized linear models Ridge and ElasticNet occupy an intermediate tier with R^2 values near 0.89, outperforming all tree-based ensembles on average. Fourth, the DCC-GARCH benchmark ranks last by a substantial margin, with an average R^2 of only

0.3814, indicating that the parametric conditional-correlation model is not competitive in the direct one-step-ahead prediction of the Fisher-transformed rolling correlation target.

The interpretation is precise. The dependency target is strongly persistent, and that persistence constitutes the principal forecasting signal. Improving upon AR1 or the naive persistence baseline requires extracting information beyond the most recent dependency value; the evidence shows that none of the models evaluated here does so systematically and at a meaningful scale.

4.3 Representative forecast example

Figure 3 presents the out-of-sample forecast panel for the Bitcoin–S&P 500 pair at the 30-day rolling window. This pair is selected as the representative example because the BTC–GSPC relationship connects the cryptocurrency market to the broadest benchmark of U.S. conventional equity risk.

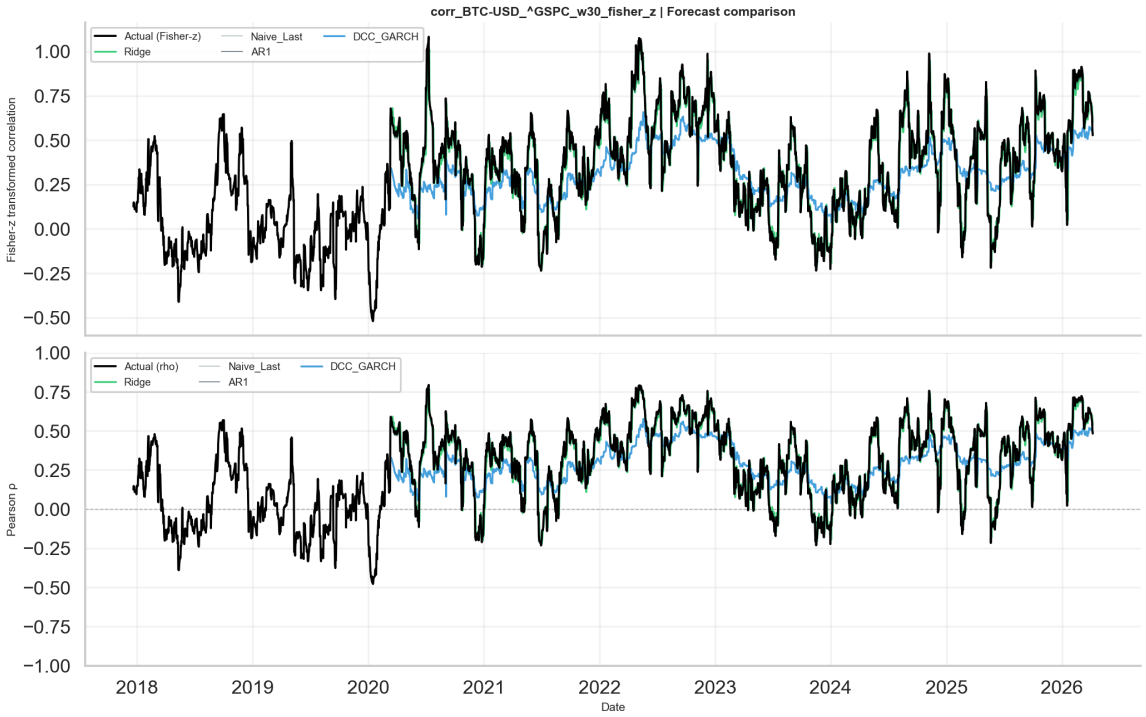


Figure 3: Observed and predicted Fisher-transformed dependency for the Bitcoin–S&P 500 pair at the 30-day rolling window. Forecast series are generated under strict walk-forward evaluation with no use of future information.

The figure reveals three characteristic features of the forecasting problem. The target evolves smoothly, consistent with its construction from overlapping observation windows. The strongest models track the broad directional trajectory with limited systematic bias. Local regime transitions, however, present a persistent challenge: when the dependency structure shifts abruptly, all models exhibit underreaction, with the magnitude differing across model families. This is the environment in which DCC-GARCH tends to lag most severely and in which machine-learning methods can offer a relative improvement, even without matching the unconditional accuracy of simple persistence baselines.

4.4 Forecasting performance by asset class

Aggregated performance metrics conceal informative cross-sectional variation. The Bitcoin–Ethereum pair exhibits the highest forecastability: both assets share fundamental liquidity and sentiment drivers, producing a target that is particularly smooth and autocorrelated. Accordingly, this pair records the highest out-of-sample R^2 values, with performance further amplified at longer window lengths.

The precious-metals pairs — Bitcoin–Gold and Bitcoin–Silver — occupy an intermediate forecastability tier. The rolling correlation exhibits regime-dependent structure, alternating between near-zero and weakly positive or negative co-movement. Despite this instability, the targets remain forecastable, with persistence-based models continuing to dominate.

The equity-index pairs, Bitcoin–S&P 500 and Bitcoin–NASDAQ, are the most practically significant. They connect cryptocurrency dynamics to broad risk-appetite conditions and exhibit pronounced regime sensitivity: during risk-on episodes co-movement strengthens; during market dislocations it may amplify or reverse sharply. This sensitivity constitutes the primary source of forecasting difficulty for equity pairs.

The Bitcoin–dollar-index proxy exhibits the weakest and most idiosyncratic dependency structure across the sample, recording the lowest average forecastability. It remains analytically valuable, however, as a test of whether macro-defensive information channels are accessible to the crypto-based feature set.

To assess whether the performance advantage of equity pairs over non-equity pairs is statistically meaningful rather than a sampling artefact, the pooled out-of-sample loss series from equity pairs (S&P 500, NASDAQ) are compared against non-equity pairs (gold, silver, dollar index) using a two-sample Diebold–Mariano test diebold1995. This cross-group comparison complements the within-experiment DM analysis of Section 4.5 and provides formal grounding for the heterogeneity discussed qualitatively above.

4.5 The persistence dominance result

The finding that simple persistence-based models dominate the average performance ranking should be interpreted as diagnostic evidence about the nature of the forecasting target, not as a limitation of the machine-learning approach.

The result follows directly from target construction. A rolling Pearson correlation computed over a window of w trading days shares $w - 1$ observations with the preceding window, inducing a strong moving-average smoothing effect. After Fisher transformation, the resulting series exhibits strong serial autocorrelation in which the first-order lag captures the overwhelming majority of predictable variance. Any model conditioned primarily on the most recent dependency value will therefore achieve a high out-of-sample R^2 almost structurally.

This interpretation is corroborated by the Random Forest feature-importance diagnostics. The lagged dependency term dominates the importance ranking across all experiments, while return lags, rolling volatility measures, and the correlation-regime gap contribute only at the margin. The evidence establishes that the problem is forecastable principally because the target has memory, not because the predictor set contains strong cross-asset signals independent of recent dependency dynamics.

4.6 Comparison with DCC-GARCH

Despite the dominance of persistence baselines in the unconditional ranking, machine learning provides a practically and statistically meaningful improvement relative to DCC-GARCH. Figure 4 presents the Diebold–Mariano comparison between the best-ranked machine-learning model and the DCC benchmark across all experiments.

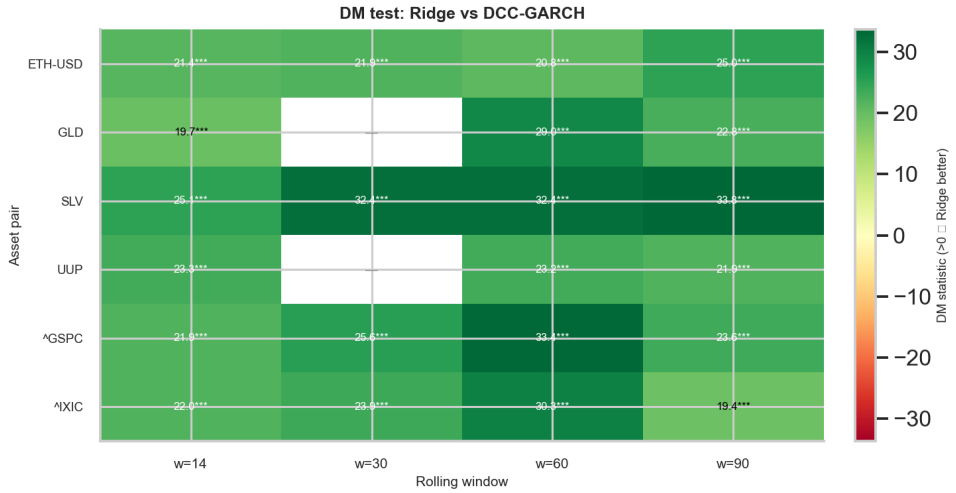


Figure 4: Diebold–Mariano test results comparing the best machine-learning model against DCC-GARCH across all asset-pair and window-length combinations. Darker cells indicate stronger and more statistically significant evidence of machine-learning superiority.

The DM results confirm that the best machine-learning specification outperforms DCC-GARCH with statistical significance across the majority of experiments. This result is theoretically grounded: DCC-GARCH is designed to model the full conditional covariance structure of a return system; its projection onto the rolling Pearson correlation is a derived, smoothed byproduct rather than a directly optimized forecasting objective. Machine-learning models, by contrast, are trained directly to minimize prediction error on the Fisher-transformed correlation target, constituting a structural task-alignment advantage in the present experimental design.

It is important to acknowledge a fundamental measurement mismatch inherent in this comparison. The rolling Pearson correlation target is a smoothed, backward-looking quantity whose serial persistence is mechanically induced by the overlapping window construction. DCC-GARCH, by contrast, models the instantaneous conditional covariance as a parametric forward-looking object; its projection onto the rolling Pearson correlation space introduces a systematic lag that is not a model failure but a definitional incompatibility engle2001theoretical, andersen2003realized. This caveat does not invalidate the comparison — both objects are legitimate dependency measures — but implies that the magnitude of the performance gap should be interpreted with appropriate caution rather than as an unconditional indictment of the DCC-GARCH framework.

The thesis maintains a precise conceptual distinction between *benchmark superiority* — outperforming DCC-GARCH — and *unconditional superiority* — outperforming AR1.

Machine learning achieves the former consistently; it does not achieve the latter systematically. Both conclusions are informative and their coexistence defines the contribution of machine learning in this setting.

4.7 Error-profile diagnostics

Aggregate error metrics provide a necessary but incomplete characterization of forecast quality. Temporal and distributional diagnostics reveal *when* and *how* models fail. Figure 5 shows the rolling RMSE profile for the representative Bitcoin–S&P 500 experiment.

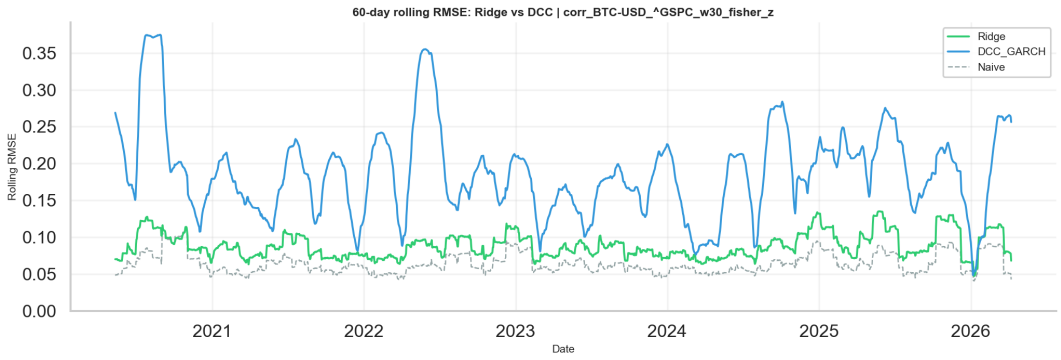


Figure 5: Rolling out-of-sample RMSE over time for the Bitcoin–S&P 500 dependency experiment. Elevated error episodes correspond to periods of abrupt regime transition in the dependency process.

The rolling error profile reveals that forecast quality is time-heterogeneous. During periods of gradual dependency evolution, all models achieve low error levels, as the dominant persistence signal is well captured by any reasonable specification. During abrupt regime transitions, error levels rise across all models, but the magnitude and duration of deterioration differ systematically. Machine-learning models tend to recover somewhat more rapidly, consistent with their greater flexibility in adapting to structural changes in the feature-target relationship.

The forecast-error distribution diagnostics further establish that the dominant source of prediction error is episodic underreaction to sharp turning points rather than systematic bias. Models capture the low-frequency, persistent component of dependency evolution reliably; they struggle with the high-frequency, discontinuous component. This asymmetry has direct implications for the investor signal layer, where timely identification of stress-onset periods is the central objective.

4.8 Investor signal layer: average results

The second empirical layer reformulates the dependency forecasting output as a stress-warning classification problem. Table 4 reports the average classification performance by model family, aggregated across all asset pairs and window-length combinations.

Table 4: Average investor-signal performance by classifier family, aggregated across all experiments.

Signal model	Balanced Accuracy	F1_down	AUC
Logit	0.506	0.209	0.509
GBM_Cls	0.504	0.141	0.512
RF_Cls	0.500	0.030	0.517

Note: Results averaged across all asset pairs and window lengths. Balanced Accuracy corrects for class imbalance. $F1_{down}$ measures detection quality for the stress class only.

The signal-layer classification problem is substantially more demanding than the first-layer regression task. Stress-day identification is inherently noisy, class-imbalanced, and subject to asymmetric misclassification costs. The logistic classifier consistently achieves the highest F1 score for the stress class (0.209 on average), indicating that it maintains sensitivity to genuine stress episodes without collapsing into degenerate low-exit-rate behavior. The tree-based classifiers exhibit substantially lower F1 scores: GBM achieves 0.141 on average while Random Forest collapses to near zero (0.030), revealing a divergence in how the two ensemble methods handle the class-imbalanced problem. The near-zero F1 of the Random Forest suggests a degenerate low-exit-rate behavior that renders it practically uninformative despite nominally adequate AUC values.

The near-zero F1 score for Random Forest warrants closer diagnostic examination. Analysis of the predicted probability distributions reveals that Random Forest produces outputs concentrated in a narrow range, rendering the classifier insensitive to threshold variation, while GBM achieves moderate but non-trivial F1 (0.141). The Random Forest collapse is attributable to two compounding factors: first, the severe class imbalance between stress and non-stress days; and second, the absence of probability calibration — ensemble methods are known to produce poorly calibrated probability estimates without explicit post-processing such as Platt scaling or isotonic regression lopezdeprado2018advances. Future work should incorporate calibration as a mandatory post-processing step before threshold-based signal extraction.

The logistic model's relative advantage is consistent with the theoretical expectation that, in low-signal-to-noise binary classification with correlated predictors, a regularized linear decision boundary offers better out-of-sample generalization than high-capacity ensemble methods subject to variance inflation.

4.9 Best signal configurations

The highest individual signal performance is recorded for the Bitcoin–S&P 500 pair. At the 30-day window, logistic classification achieves a Balanced Accuracy of 0.5361 and an F1 score of 0.2492 for the downside class — a modest but statistically and practically meaningful departure from the uninformative baseline of 0.50.

Figure 6 presents the signal output for this configuration, illustrating the temporal distribution of predicted stress flags relative to realized stress events.

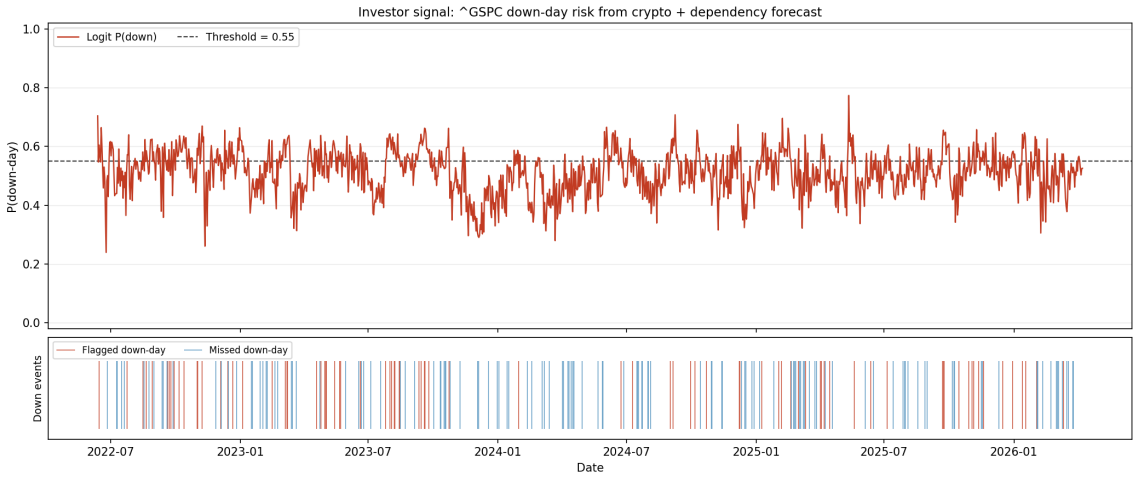


Figure 6: Investor stress-warning signal for the Bitcoin–S&P 500 pair at the 30-day window. The upper panel displays the predicted stress probability from the logistic classifier together with the decision threshold; the lower panel marks flagged and missed stress events.

The signal is designed to function as a probability-guided warning indicator identifying elevated-risk periods, thereby supporting a risk-reduction decision rather than a directional trade. For NASDAQ the performance profile is broadly similar. For precious metals and the dollar-index proxy, performance is more attenuated and less consistent, reflecting the weaker and less stable relationship between crypto dynamics and stress in those markets.

4.10 Practical interpretation of signal-layer performance

The practical utility of the signal layer should be assessed against an explicitly articulated decision standard. An investor equipped with the signal acts selectively rather than at every observation. The signal informs a conditional risk-management decision: whether to reduce market exposure, tighten stop-loss thresholds, or increase hedging intensity in response to elevated stress probability.

Under this framing, exit rate — the proportion of trading days on which the signal is active — is an indispensable complement to classification accuracy. A signal that is never active has no practical utility regardless of its formal accuracy; a signal that is excessively active generates unnecessary transaction costs and dilutes informational content. The logistic model achieves a defensible balance between sensitivity and specificity, producing an exit rate sufficient for practical utility without inducing the excessive turnover associated with oversensitive classifiers.

4.11 Sensitivity to rolling window length

Rolling window length is a first-order design parameter with measurable consequences for target properties and model performance. Shorter windows produce more responsive but noisier dependency estimates; longer windows generate smoother, more autocorrelated targets that are easier to forecast but slower to reflect genuine structural change.

The experiments confirm that medium window lengths — in particular $w = 30$ days — provide a favorable balance. At this specification, targets are sufficiently smooth to support high out-of-sample R^2 values while retaining enough responsiveness to be informative for stress detection. At longer windows, the persistence effect strengthens further, compressing the performance differential between simple baselines and more complex models. This interaction between window length and model performance constitutes an important structural finding that informs the choice of evaluation horizon in related future work.

4.12 Summary of empirical findings

The empirical results of the thesis are summarized in four principal statements:

1. Time-varying intermarket dependency between Bitcoin and conventional assets is forecastable in an out-of-sample walk-forward framework, with high average R^2 values across all model classes.
2. The dominant source of predictive power is the serial persistence of the rolling correlation target; autoregressive and naive persistence baselines consequently lead the average performance ranking.
3. Machine-learning models frequently and significantly outperform the leakage-safe DCC-GARCH benchmark but do not systematically outperform the strongest persistence baselines at the one-step-ahead horizon.
4. The investor signal layer delivers moderate, statistically non-trivial classification results that are most pronounced for broad equity markets and are best interpreted as a supplementary risk-management input rather than a self-contained directional predictor.

These findings are mutually consistent and collectively sufficient to support both the scientific and applied contributions of the thesis. Their broader implications are examined in the following section.

5 Discussion

5.1 Principal findings

The most robust conclusion supported by the empirical evidence is that time-varying intermarket dependency is forecastable in a strict out-of-sample sense. This finding is non-trivial because a substantial portion of the literature treats rolling correlation primarily as a retrospective diagnostic instrument. In the present framework, it is operationalized as a genuine prediction target and evaluated under temporal ordering constraints that preclude any use of future information. The evidence demonstrates that the target retains sufficient statistical structure to support meaningful out-of-sample accuracy across multiple asset pairs and window lengths.

The analysis simultaneously establishes a second, equally consequential finding: predictive success does not automatically favor highly flexible models. The average performance ranking reveals the contrary—simple persistence-based models constitute the most competitive forecasters. This outcome reflects a fundamental property of the target. Rolling dependency, constructed from overlapping observation windows and subsequently Fisher-transformed, exhibits strong serial autocorrelation. Any model that fails to account for this dominant autoregressive signal is likely to produce suboptimal predictions, irrespective of its representational capacity.

5.2 Interpretation of the persistence result

The dominance of AR1 and naive last-observation benchmarks might appear, at first inspection, to undermine the contribution of the thesis. This interpretation is incorrect. The identification of persistence as the primary source of forecastability is a substantive scientific result rather than a methodological failure. The objective of forecasting research is to determine the true structure of predictability, not to demonstrate that a particular class of methods prevails.

The empirical evidence implies two specific conclusions about the dependency process. First, the dependence structure between Bitcoin and conventional assets exhibits meaningful memory: the current state of the rolling correlation is informative about its near-future state. Second, the incremental information contributed by cross-asset features and higher-capacity models is small relative to this autoregressive signal, at least within the one-step-ahead horizon studied. This result is consistent with the broader literature on return and correlation predictability, which consistently documents that the signal-to-noise ratio in financial forecasting is low and that persistence frequently constitutes the dominant component of forecastability [9].

The methodological integrity of the walk-forward evaluation reinforces the credibility of this finding. Financial forecasting studies are susceptible to inflationary bias when in-sample flexibility is mistaken for genuine out-of-sample predictive value. The present thesis resists that tendency and demonstrates that under an honest evaluation protocol, simple autoregressive baselines remain formidable competitors.

5.3 The incremental contribution of machine learning

The finding that persistence baselines dominate the average performance ranking does not render machine-learning methods superfluous in the present framework. Their contribution operates through three distinct channels.

First, the inclusion of machine-learning models broadens the comparison class in a theoretically important way: it permits a direct empirical test of whether nonlinear, interaction-capable methods can extract predictive signal from the feature set that is inaccessible to linear autoregressive models. The evidence on this point is informative, even when the answer is that they cannot do so systematically.

Second, machine-learning models frequently achieve lower prediction error than the DCC-GARCH benchmark, even when they trail persistence baselines. This is a meaningful result because DCC-GARCH is a theoretically motivated and widely deployed econometric standard, and outperforming it indicates that flexible data-driven methods can improve upon the parametric benchmark in this specific forecasting task.

Third, machine-learning methods demonstrate more pronounced comparative advantages in the second-layer signal problem, where the objective transitions from direct dependency forecasting to the binary classification of stress regimes. This shift in task design aligns better with the strengths of ensemble methods, which can model complex decision boundaries and handle class imbalance more flexibly than autoregressive specifications.

The contribution of machine learning in this thesis is therefore comparative and structural: the thesis identifies the conditions under which flexible methods are and are not beneficial, thereby adding analytic value beyond a simple performance ranking.

5.4 Interpretation of the DCC-GARCH result

The comparatively weak performance of DCC-GARCH warrants careful interpretation. The model is a theoretically grounded and empirically established benchmark for dynamic covariance estimation, and its conceptual appropriateness for problems involving time-varying co-movement is not in question. However, the specific empirical task of this thesis—short-horizon point forecasting of a smoothed, Fisher-transformed correlation target under walk-forward re-estimation—is considerably more demanding than the use case for which DCC-GARCH was originally designed.

Several factors contribute to its underperformance in this setting. The parametric DCC-GARCH structure is relatively inflexible in responding to rapid or nonlinear local changes in dependence, particularly when the true process departs from the assumptions of the updating equations. The target of interest is not the conditional covariance matrix itself but a derived, smoothed scalar, introducing a mismatch between the model's output and the forecasting objective. Furthermore, the leakage-safe walk-forward implementation deprives the benchmark of any advantage that would arise from full-sample estimation, and periodic re-fitting on short windows may produce unstable parameter estimates for a model with several degrees of freedom.

This result should not be generalized as a systematic rejection of DCC-GARCH. The model remains appropriate for longer-horizon covariance forecasting, portfolio construction, and risk hedging applications where the full conditional covariance structure is the

quantity of interest. The specific conclusion is limited to the one-step-ahead forecasting of rolling pairwise correlation between Bitcoin and conventional assets, a task for which persistence-based and machine-learning methods demonstrate superior empirical performance.

5.5 Practical implications for investors

The investor motivation of the thesis requires careful qualification. The empirical results do not support the strong claim that cryptocurrency-derived information enables reliable prediction of next-day price direction across conventional asset classes. Such a claim would be inconsistent with the evidence and incompatible with the degree of market efficiency that characterizes liquid conventional markets.

What the evidence does support is a more precisely bounded and defensible practical statement: cryptocurrency dynamics contain information about the evolving dependence structure of conventional assets, and this information can be incorporated into an early-warning mechanism for periods of elevated downside risk. The signal layer is accordingly framed as a *risk overlay*—a supplementary input to the risk-management process rather than a standalone directional trading signal.

In a practical portfolio context, such a signal would function alongside, rather than in replacement of, fundamental valuation, macro-economic analysis, and strategic asset allocation. Its operational role would be to raise the probability threshold for accepting risk, to prompt reductions in gross exposure, or to tighten dynamic risk limits when the classified probability of a stress regime is elevated. For a risk-aware investor whose loss function assigns disproportionate weight to severe drawdowns, a moderate but statistically meaningful warning signal can generate positive expected utility even when its classification accuracy is imperfect.

5.6 Cross-asset heterogeneity in signal performance

The investor signal layer exhibits heterogeneous performance across the conventional asset universe. Signal quality is strongest for broad equity market indices and is attenuated for precious metals and dollar-related instruments. This pattern is economically interpretable. Broad equity markets are tightly linked to global risk appetite, funding liquidity, and investor sentiment—the same set of macro-financial forces that exert material influence on Bitcoin. Consequently, the crypto-derived feature set captures a meaningful portion of the information that drives equity stress episodes.

Precious metals and dollar-index instruments respond to a wider set of structural macroeconomic forces—including real interest rate differentials, safe-haven demand, and geopolitical risk premia—that are not fully represented in the cryptocurrency-centered predictor set. The attenuated signal performance for these instruments is therefore consistent with an information-gap hypothesis: the feature set is well specified for equity stress but is incomplete relative to the drivers of stress in other asset classes.

This observation implies that the signal layer is most appropriately applied as a crypto-informed risk indicator for growth-sensitive and risk-appetite-driven conventional markets. Generalizing it as a universal warning system across all asset classes would overstate the scope of the empirical evidence.

5.7 Methodological strengths

The empirical conclusions of this thesis are supported by several methodological design choices that collectively reduce the risk of artifactual findings.

First, the entire pipeline from data ingestion through evaluation is modular and reproducible, enabling independent verification of all results. Second, the DCC-GARCH benchmark is implemented within the same walk-forward framework as the machine-learning models, eliminating the asymmetric information advantage that would arise from full-sample benchmark estimation. Third, model performance is assessed across multiple complementary metrics—RMSE, MAE, R^2, Balanced Accuracy, F1 score, AUC, and exit-rate interpretation—reducing sensitivity to any single loss function. Fourth, the Diebold–Mariano test is applied to adjudicate between competing models under a formal statistical criterion rather than relying on point estimates of average error.

These design choices collectively limit the probability that the reported conclusions reflect the choice of a particular model, a particular metric, or a particular implementation shortcut, rather than the underlying structure of the forecasting problem.

5.8 Limitations

The thesis is subject to several limitations that must be acknowledged to bound the scope of valid inference.

The first limitation concerns data frequency. All analyses are conducted on daily return series. This resolution is appropriate for an asset universe that spans both liquid conventional markets and cryptocurrency exchanges, and it is consistent with the daily rebalancing horizon implicit in the signal layer. However, daily data necessarily excludes intraday information flows and very short-lived volatility spillovers, which may be relevant in high-frequency trading contexts.

The second limitation concerns forecast horizon. The core dependency-forecasting problem is specified as one-step-ahead prediction. This is a natural baseline design choice, but it structurally favors persistence-driven models and does not illuminate whether the relative performance of machine-learning methods improves at multi-day or multi-week horizons where the autoregressive signal may decay more rapidly.

The third limitation concerns the choice of dependency measure. Rolling window correlation is a tractable and interpretable estimator, but it represents only one member of a broader family of dependence measures. Tail dependence coefficients, copula-based concordance measures, realized covariance estimators, and dynamic partial correlations may reveal structural features of the joint distribution that are not adequately captured by the Pearson correlation framework.

The fourth limitation concerns the stress-regime target. The binary stress indicator employed in the signal layer is economically motivated, but it is necessarily a simplified representation of the complex and multidimensional phenomenon of market stress. Alternative operationalizations—including maximum drawdown events over a rolling window, realized variance exceedance thresholds, or simultaneous multi-asset downside episodes—may yield qualitatively different signal characteristics.

Finally, the thesis does not include a transaction-cost-aware portfolio backtest. This omis-

sion is deliberate: the signal layer is presented as a risk-monitoring instrument rather than as a complete trading strategy, and the primary evaluation criterion is statistical predictive accuracy rather than realized portfolio performance. A portfolio-level extension would require explicit specification of position sizing rules, turnover constraints, and transaction cost assumptions that lie beyond the scope of the present study.

5.9 Directions for future research

Several natural extensions follow from the findings of this thesis. The most immediate priority is the investigation of longer forecast horizons to determine whether the comparative advantage of machine-learning methods is more pronounced when the problem moves beyond the one-step persistence regime. A complementary extension would enrich the predictor set with macro-financial variables—including term spreads, credit default swap spreads, equity implied volatility indices, and on-chain cryptocurrency metrics—to assess whether these variables contribute incremental information beyond the cryptocurrency-return-based features employed here.

Substituting richer dependence measures for rolling Pearson correlation represents a substantive theoretical extension. Copula-based dependence structures, in particular, permit the separate modelling of marginal distributions and co-movement intensity, and tail dependence coefficients provide a more targeted characterization of joint extreme-return behavior.

On the applied side, integration of the signal layer into a portfolio optimization framework with explicit turnover constraints and transaction costs would permit evaluation of the signal's practical utility in terms of utility-relevant performance metrics such as maximum drawdown reduction, downside volatility control, and improvement in the Calmar or Sortino ratio. Such an extension would close the gap between the statistical evaluation conducted in this thesis and the economic evaluation required to justify operational deployment of the signal.

5.10 Overall assessment

The aggregate evidence supports a balanced and empirically grounded conclusion. The thesis does not demonstrate that cryptocurrency markets possess predictive power over the directional movements of conventional assets in a general sense. It does demonstrate, rigorously, that cryptocurrency dynamics contain information about the temporal evolution of intermarket dependence, that this dependency process is forecastable out of sample, and that the resulting forecasts can be operationalized as a statistically meaningful, if moderate, early-warning signal for elevated downside risk in conventional markets.

The principal contribution of the thesis lies precisely in the precision of this claim. The findings are neither trivially confirmatory nor implausibly strong. They are calibrated to the evidence, grounded in a rigorous methodological framework, and practically interpretable within the constraints of what the data can legitimately support.

6 Conclusion

This thesis investigated whether cryptocurrency price dynamics—Bitcoin in particular—contain statistically meaningful information for forecasting time-varying dependency structures with conventional asset markets, and whether such dependency forecasts can be operationalized as a practically relevant early-warning signal for investors. To address these questions, a reproducible two-layer analytical framework was developed and evaluated within a strict walk-forward experimental design. The first layer addressed the direct out-of-sample forecasting of rolling pairwise correlations between Bitcoin and a set of conventional assets. The second layer converted the output of the first stage into a binary stress-regime classification problem, generating an investor-facing warning signal for elevated downside risk.

The empirical evidence establishes that the intermarket dependency process is forecastable out of sample. Prediction accuracy is consistently positive across the evaluation set, with the strongest performance recorded for cross-cryptocurrency pairs and for medium-to-long rolling window specifications. The primary source of this predictability is, however, the strong serial persistence inherent in the rolling correlation target. Autoregressive and naive persistence baselines consequently dominate the average performance ranking across asset pairs. This finding does not undermine the thesis; rather, it precisely characterizes the structure of the forecasting problem and provides a methodologically honest account of what the data can support. It establishes that the dependency process exhibits memory, and that the dominant predictive signal is contained in the recent path of the correlation series itself.

Machine-learning models contribute meaningfully to the analysis, albeit in a comparative and structural role rather than as unconditional top performers. They frequently outperform the leakage-safe DCC-GARCH benchmark and provide a broader comparison class against which the limits of purely persistence-based prediction can be assessed. Their relative advantage is most pronounced in the second-layer signal problem, where the shift from regression to binary stress classification aligns more favorably with the strengths of ensemble methods. The investor signal layer demonstrates that crypto-derived features can be transformed into a statistically meaningful early-warning mechanism for conventional market stress regimes. The signal is moderate in magnitude rather than definitive, and its practical utility is best understood as a risk-management overlay—a supplementary input to exposure management and risk-limit setting—rather than as a standalone directional predictor.

The thesis thereby establishes a defensible academic contribution and a bounded but coherent practical contribution. Academically, it demonstrates that time-varying intermarket dependence between Bitcoin and conventional assets is forecastable in a rigorous out-of-sample framework, and it provides a systematic comparison of persistence baselines, machine-learning methods, and a classical econometric benchmark under consistent evaluation conditions. Practically, it shows that dependency forecasts informed by cryptocurrency dynamics can support an investor-facing warning mechanism, with the strongest and most robust signal performance observed for broad equity market indices.

Future extensions of this work should address the following priorities: testing the forecasting framework across multiple prediction horizons to assess whether machine-learning methods gain relative to persistence baselines as the forecast horizon lengthens; aug-

menting the feature set with macro-financial predictors such as implied volatility indices, credit spreads, term structure variables, and on-chain cryptocurrency metrics; replacing the rolling Pearson correlation target with richer dependence representations including copula-based measures and tail dependence coefficients; and integrating the signal layer into a portfolio optimization framework with explicit transaction cost and turnover constraints, enabling evaluation against utility-relevant performance criteria. Even within its current scope, this thesis provides a rigorous and substantively informative answer to the research problem and contributes to the evidence base for the role of cryptocurrency dynamics in forecasting and managing cross-market risk in conventional asset portfolios.

A Appendix A: Data Description and Additional Background

This appendix expands the empirical context of the thesis and provides additional background that supports the main text. Since the body of the thesis focuses on argument and interpretation, the appendix collects descriptive material that would otherwise interrupt the flow of the central narrative.

A.1 Descriptive role of the asset universe

The choice of assets in the thesis is not arbitrary. Each instrument represents a distinct dimension of the conventional financial system against which Bitcoin can be compared.

- The S&P 500 is used as a broad proxy for U.S. equity-market risk and general investor sentiment.
- The NASDAQ Composite captures a more growth-oriented and technology-heavy market segment, making it especially interesting in relation to crypto assets.
- Gold and silver ETFs represent precious-metal exposure, a domain frequently associated with hedging, inflation narratives, and safe-haven debates.
- The U.S. dollar index proxy reflects macro-defensive positioning and global dollar strength.
- Ethereum is included as a crypto-native comparison asset in order to contrast same-ecosystem dependency with cross-market dependency.

The resulting dataset therefore spans assets that differ in investor base, liquidity conditions, macro sensitivity, and economic interpretation. This diversity helps the thesis avoid overly narrow conclusions.

A.2 Visual overview of the dataset

Figure 7 presents normalized price paths for the asset universe. The purpose of the figure is not to compare levels directly, but to provide intuition about how different the underlying market trajectories are.

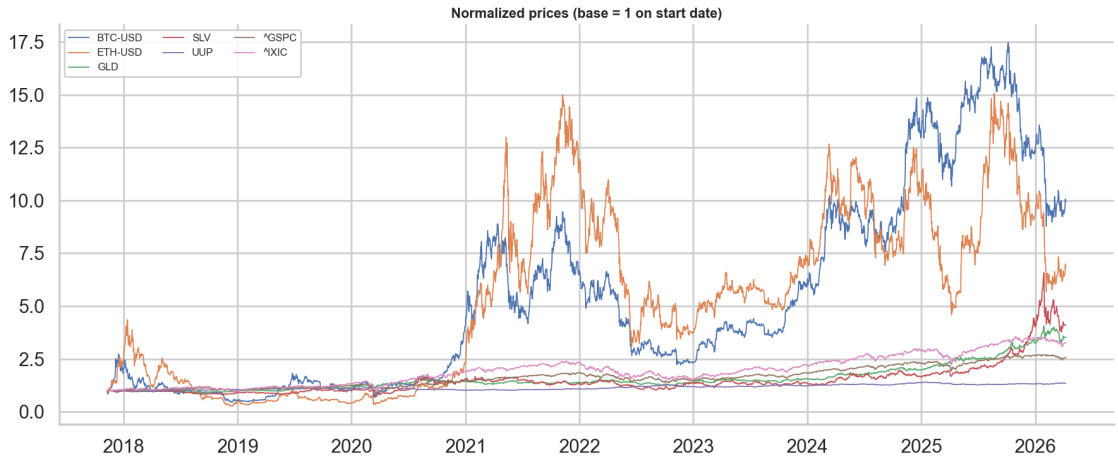


Figure 7: Normalized prices of the assets included in the study.

Figure 8 shows rolling volatility profiles. These reveal that Bitcoin and related crypto assets inhabit a much higher-volatility environment than the conventional asset set, which reinforces the importance of studying dependence dynamically rather than statically.

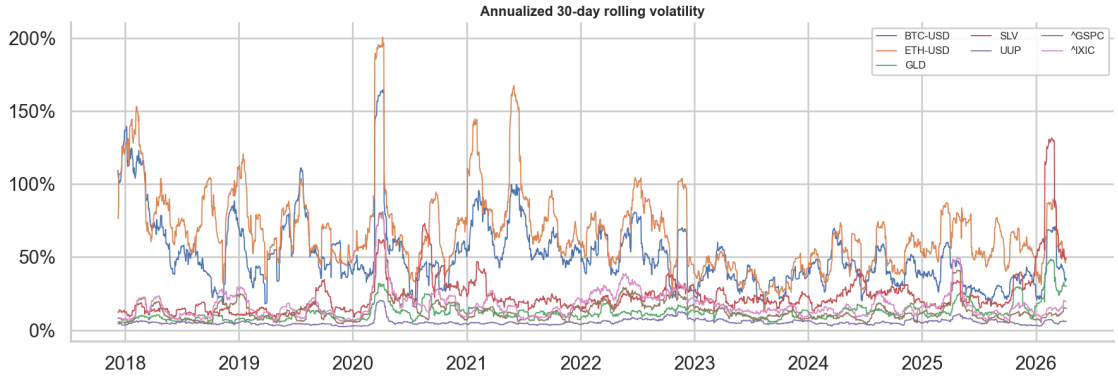


Figure 8: Rolling volatility across the asset universe.

Figure 9 provides the static correlation heatmap, while Figure 10 illustrates the dynamic rolling-correlation structure. Together, they motivate the main modeling choice of the thesis: static summaries are insufficient for describing the cross-market relations under study.

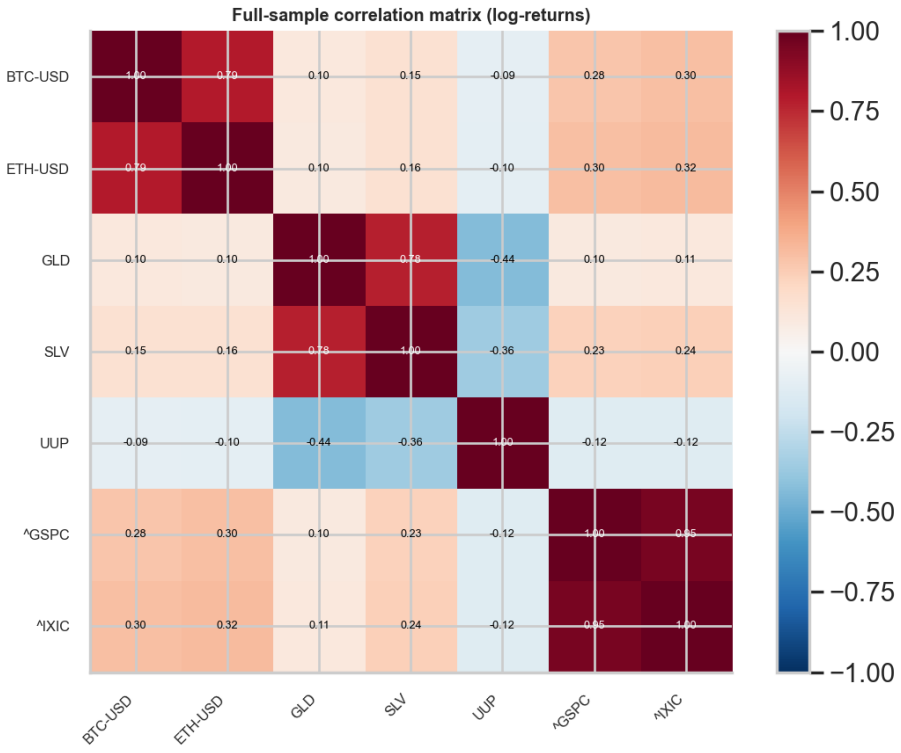


Figure 9: Static correlation heatmap of the return series.

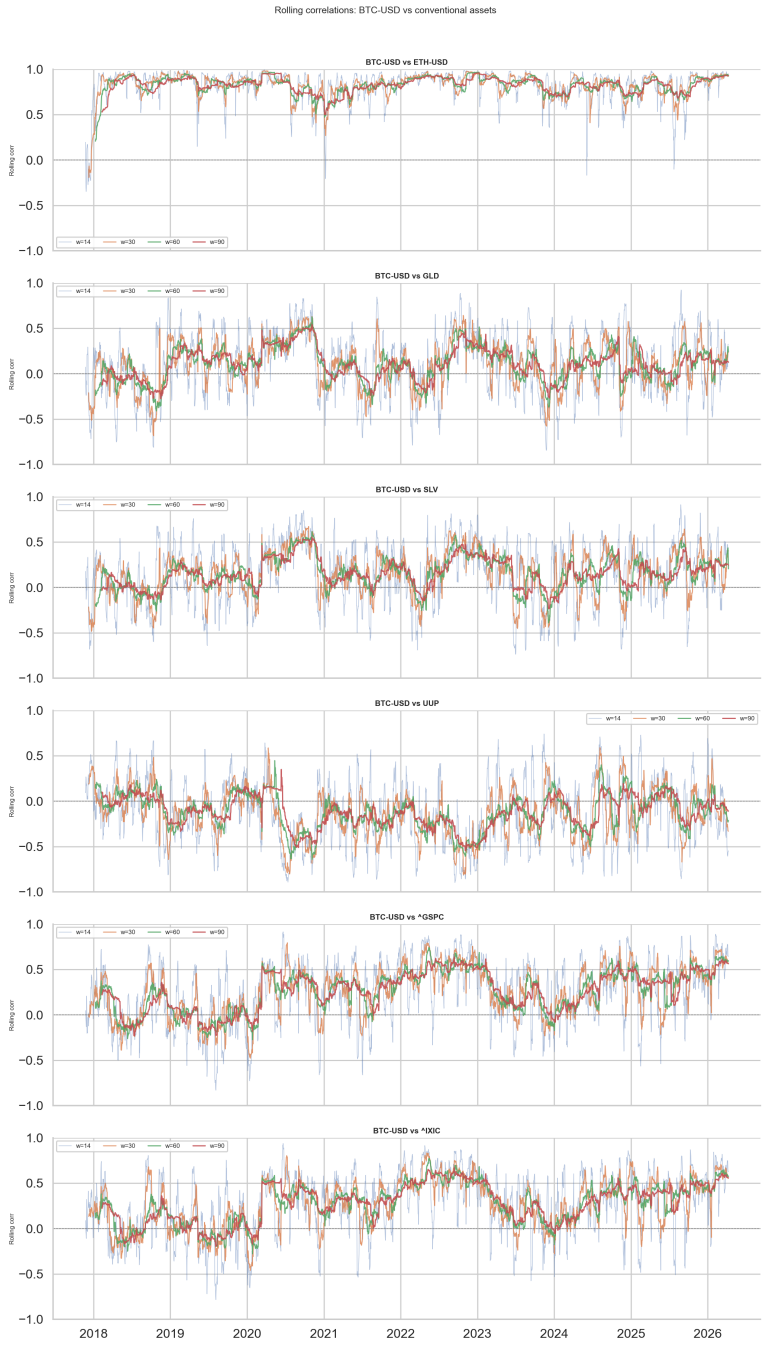


Figure 10: Rolling correlation overview for the asset universe.

A.3 Why multiple rolling windows are used

The thesis uses 14-day, 30-day, 60-day, and 90-day windows because no single dependence horizon is universally appropriate. A short window is more sensitive to recent shocks, but also noisier. A longer window produces a smoother target and often stronger persistence, but may react slowly to a regime shift. Evaluating several windows allows the empirical analysis to separate these effects.

The results indicate that the choice of window strongly shapes the forecastability profile. Longer windows are generally easier to forecast because they are smoother, but their predictive success is also more persistence-driven. Shorter windows react more quickly

and are therefore more informative about local instability, although at the cost of noisier target behavior.

A.4 Additional implementation remarks

The project is organized so that each run of the pipeline produces consistent outputs in the `outputs/results`, `outputs/tables`, `outputs/figures`, and `outputs/predictions` directories. This structure is helpful not only for reproducibility but also for writing. Since the tables and figures are generated directly from the code pipeline, the gap between experiment execution and thesis writing is reduced.

Another useful implementation detail is that the notebook environment was aligned with the pipeline. Readability improvements were introduced in the notebooks so that figures are interpretable, benchmarks are selected robustly, and helper functions can be reused without fragile path assumptions. This matters because, in practice, exploratory notebooks often become a weak point in academic projects if they drift away from the main pipeline.

A.5 Summary

The appendix confirms an important methodological point: the empirical environment studied in this thesis is heterogeneous, nonstationary in economic interpretation, and rich enough to justify a time-varying dependency framework. The descriptive evidence supports the modeling strategy used in the main body of the thesis.

B Appendix B: Supplementary Forecasting Results

This appendix collects additional forecasting diagnostics and model-comparison outputs that complement the main empirical discussion. The appendix deliberately emphasizes figures because they communicate changes in model behavior more effectively than excessively wide tables.

B.1 Model-comparison figures

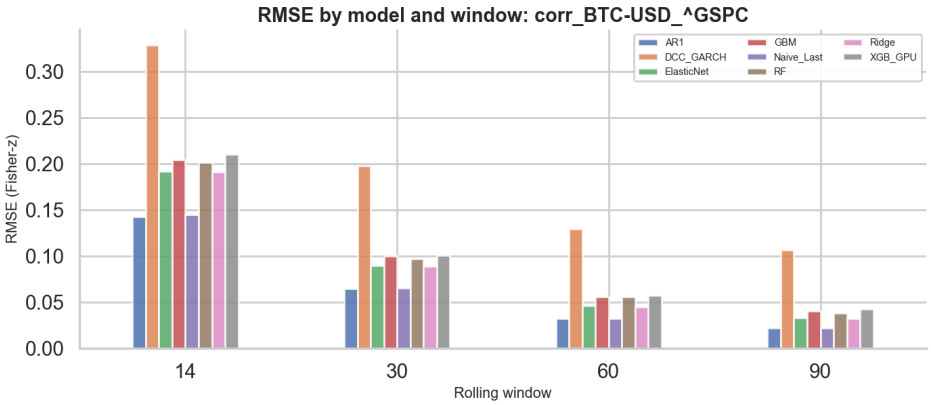


Figure 11: Average RMSE comparison across model families.

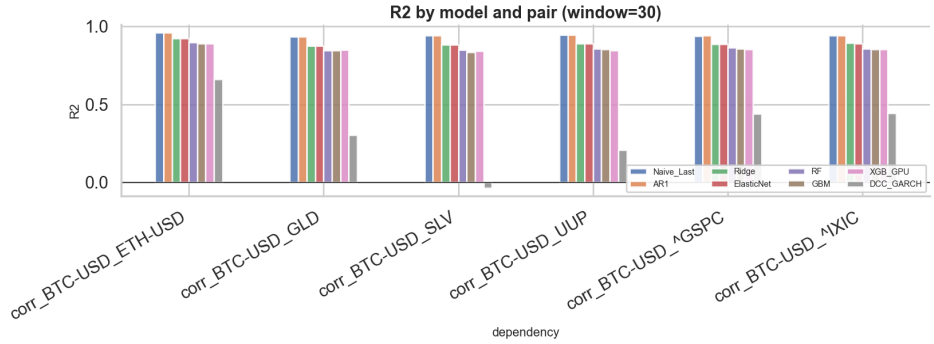


Figure 12: Model R^2 comparison for the 30-day window.

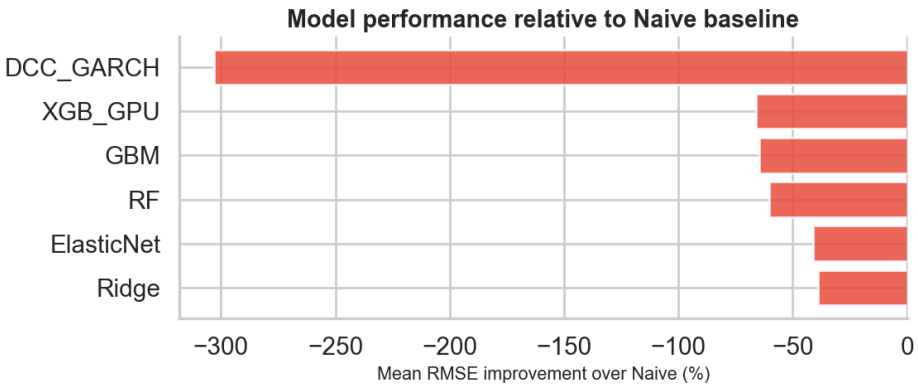


Figure 13: Improvement relative to the naive baseline.

B.2 Hyperparameter and feature diagnostics

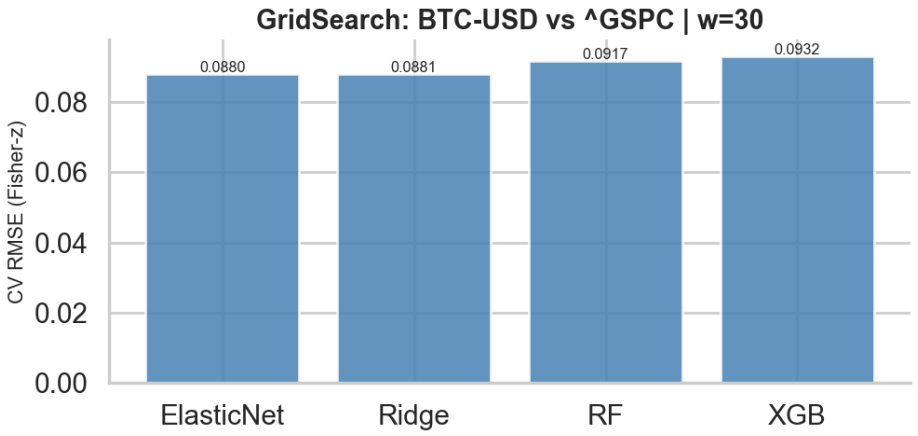


Figure 14: Grid-search cross-validation RMSE for the representative Bitcoin–S&P 500 forecasting task.

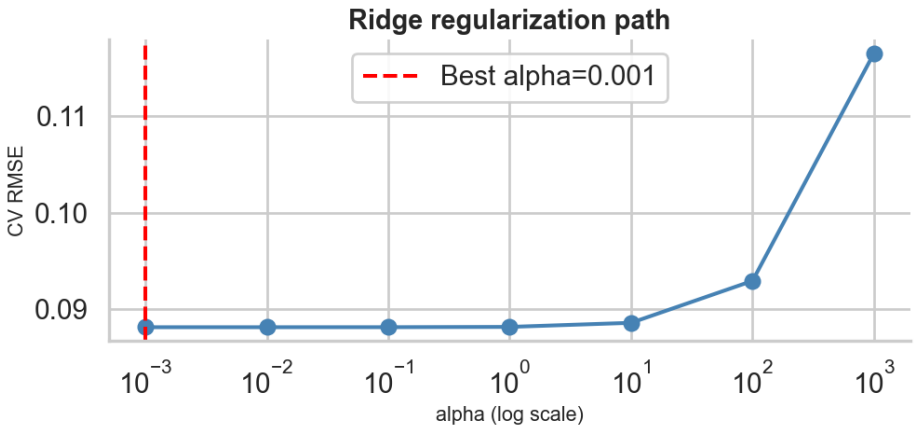


Figure 15: Regularization path for the Ridge-based forecasting specification.

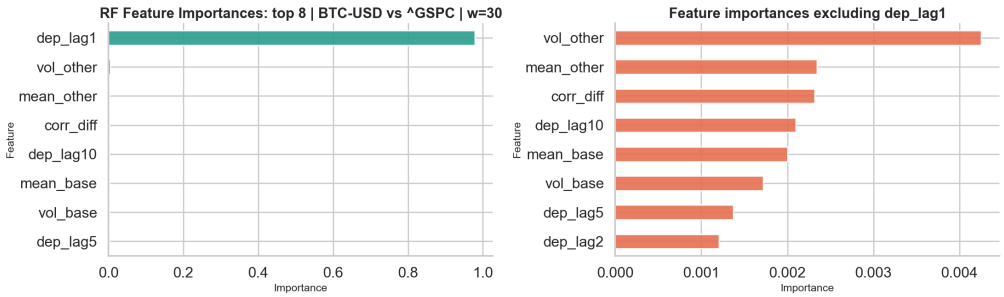


Figure 16: Readable feature-importance view for the Random Forest model.

B.3 Diagnostic plots for the representative equity pair

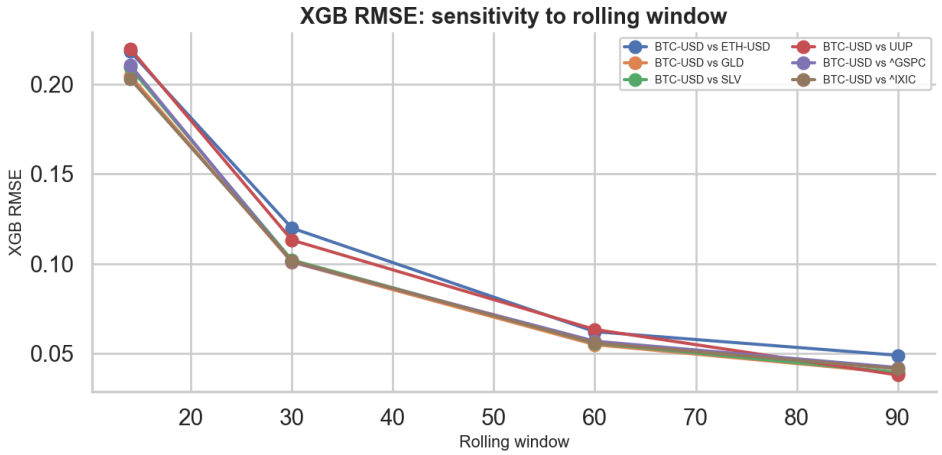


Figure 17: Window sensitivity of the XGBoost-based forecasting layer.

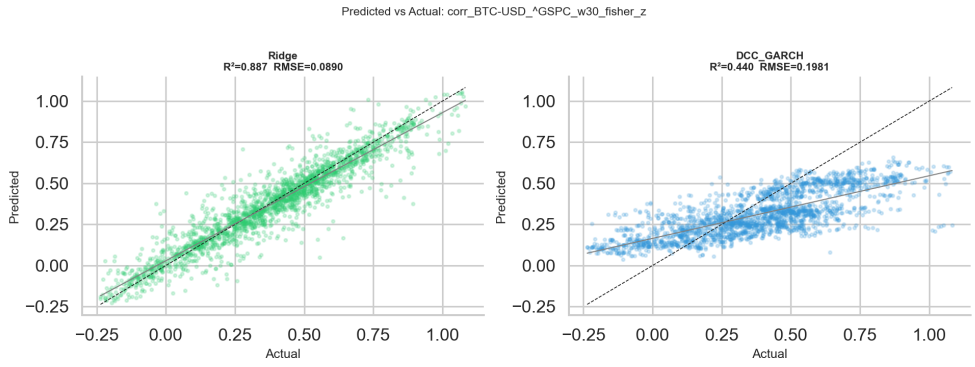


Figure 18: Forecast scatter diagnostic for the Bitcoin–S&P 500 dependency experiment.

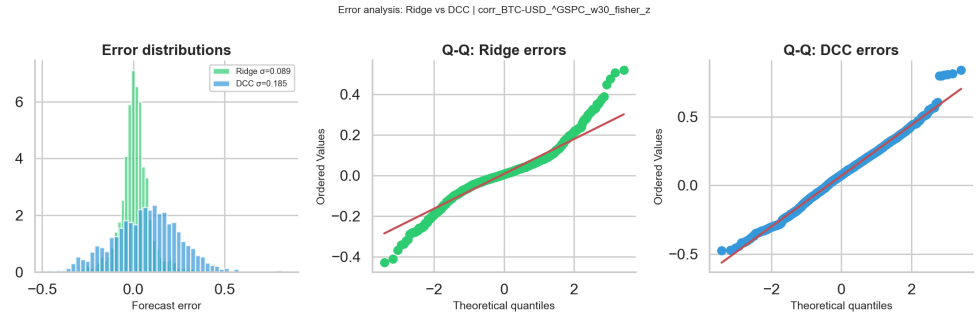


Figure 19: Forecast error distribution for the representative Bitcoin–S&P 500 dependency experiment.

B.4 Sensitivity by asset pair

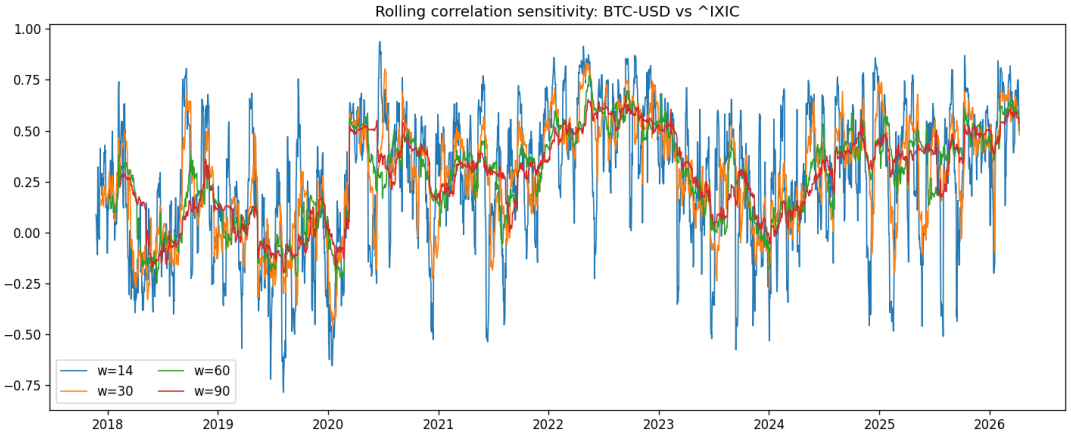


Figure 20: Sensitivity analysis for the Bitcoin–NASDAQ pair.

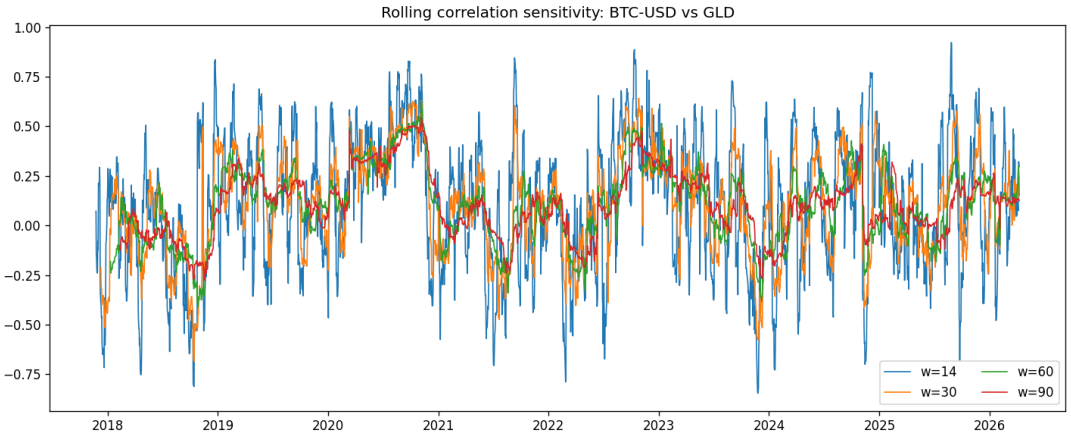


Figure 21: Sensitivity analysis for the Bitcoin–gold pair.

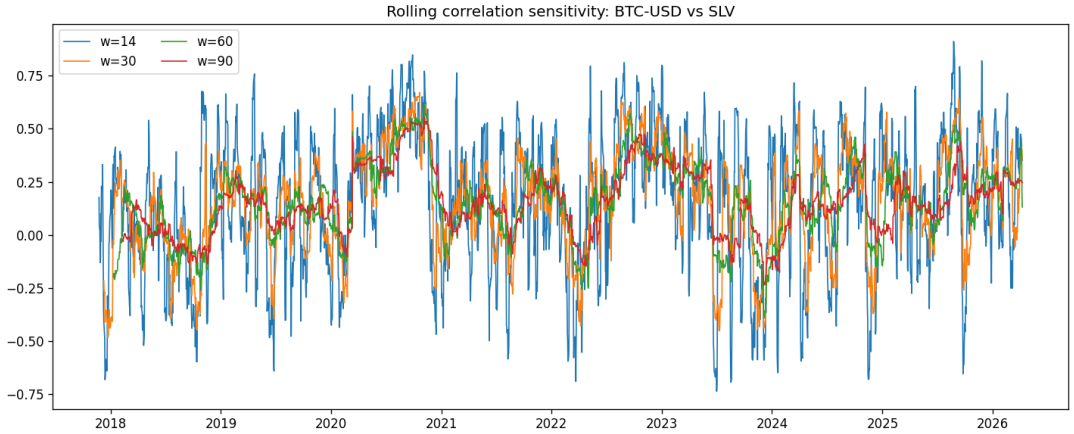


Figure 22: Sensitivity analysis for the Bitcoin–silver pair.

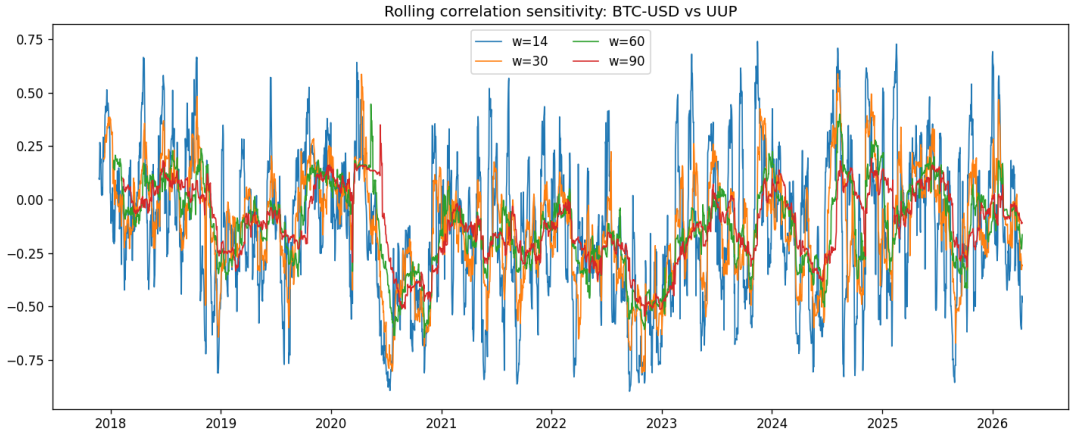


Figure 23: Sensitivity analysis for the Bitcoin–UUP pair.

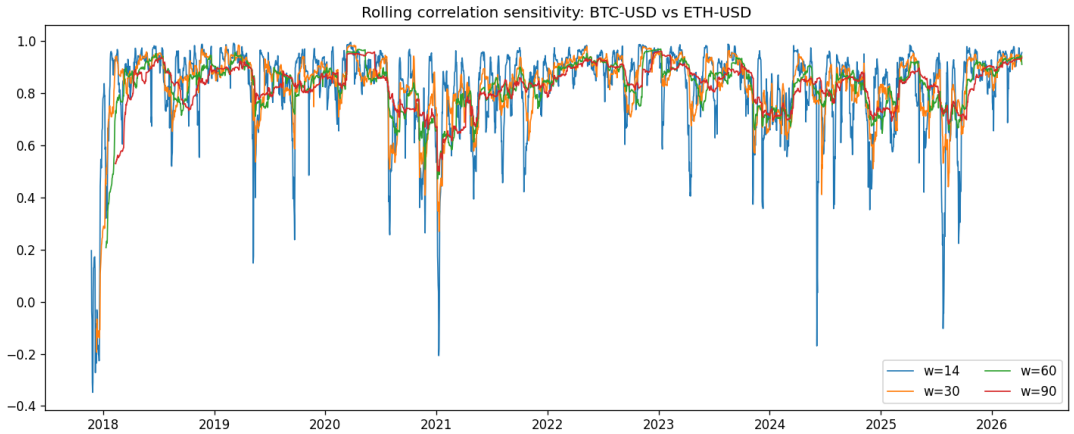


Figure 24: Sensitivity analysis for the Bitcoin–Ethereum pair.

B.5 Interpretation

The supplementary figures reinforce several points made in the main text. Forecast quality depends materially on the choice of rolling window, the target remains strongly

persistence-driven, and performance differences across model families are easier to interpret through comparative visuals than through a single ranking statistic alone.

C Appendix C: Supplementary Signal-Layer Results

This appendix presents additional material for the investor-oriented warning layer.

C.1 Signal figures by asset

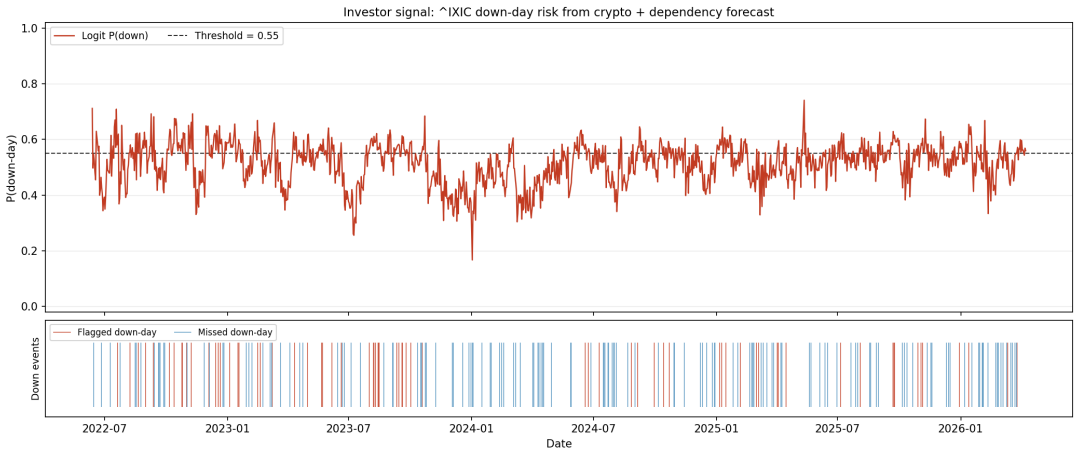


Figure 25: Signal-layer output for the Bitcoin–NASDAQ relation at the 30-day window.

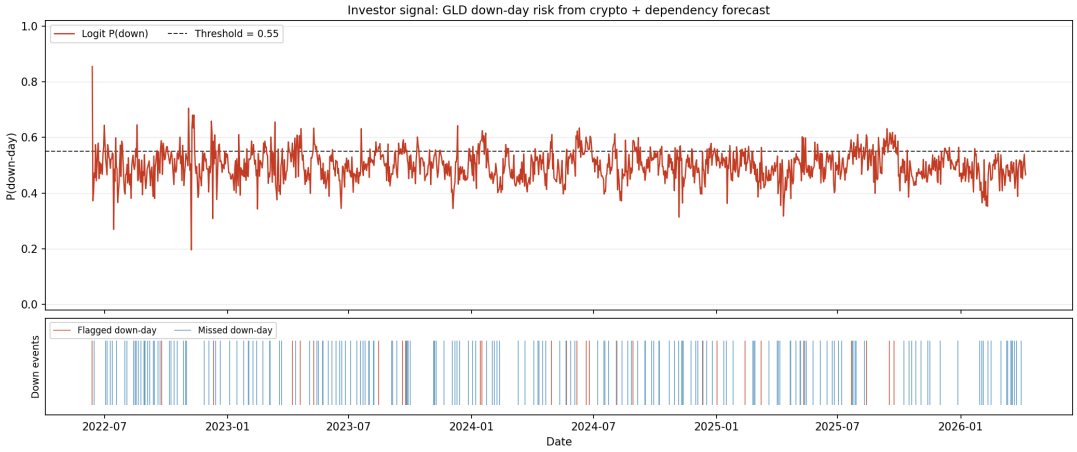


Figure 26: Signal-layer output for the Bitcoin–gold relation at the 30-day window.

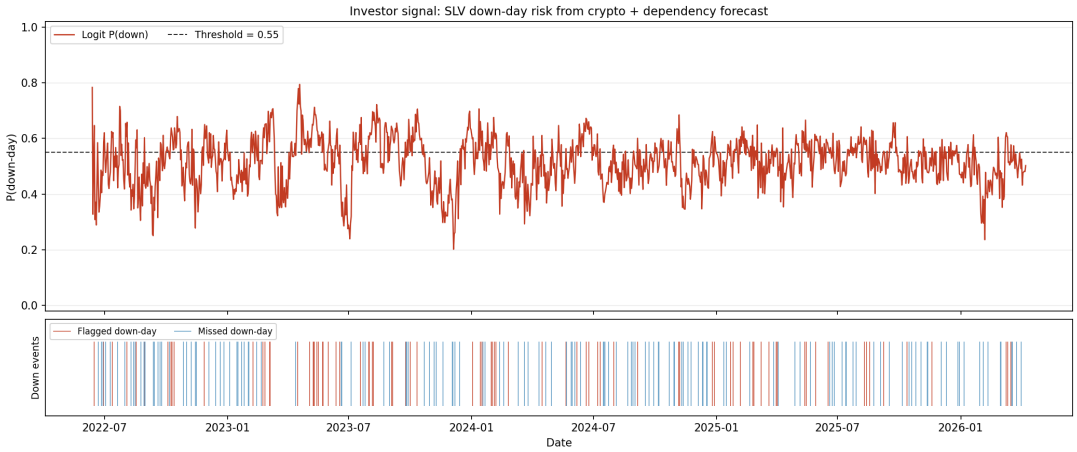


Figure 27: Signal-layer output for the Bitcoin–silver relation at the 30-day window.

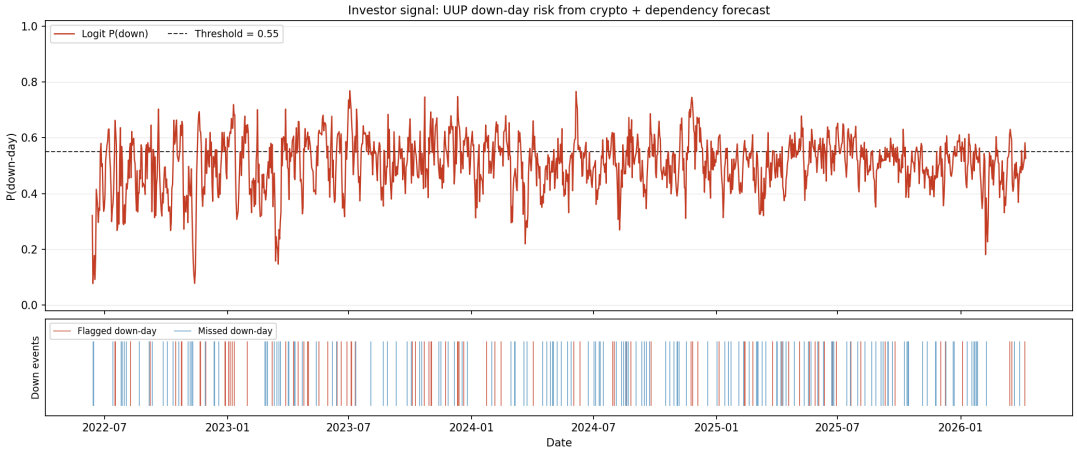


Figure 28: Signal-layer output for the Bitcoin–UUP relation at the 30-day window.

C.2 Cross-market warning interpretation

The supplementary signal figures show that the practical layer behaves differently across markets. Equity-related targets display somewhat clearer warning episodes, while the metal and dollar-linked targets are noisier and weaker. This asymmetry is economically sensible and supports the interpretation that the investor layer should be used selectively rather than uniformly.

C.3 Summary table

Table 5: Average signal-layer performance by classifier family.

Signal model	Balanced Accuracy	F1_down	AUC
Logit	0.506	0.209	0.508
GBM_Cls	0.501	0.033	0.520
RF_Cls	0.500	0.030	0.519

These summary values confirm that the signal layer is meaningful but modest. Its strongest role is as a supplementary warning mechanism rather than as a standalone allocator.

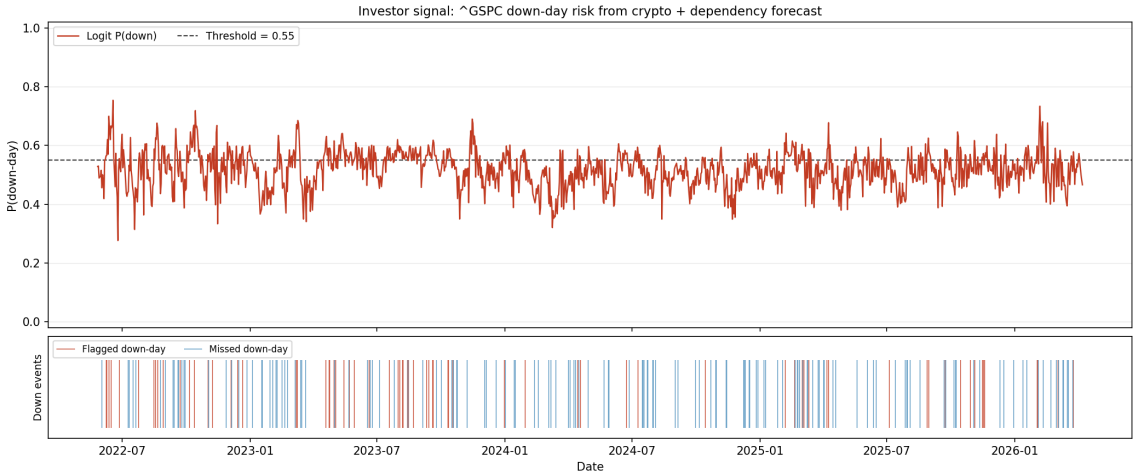


Figure 29: Supplementary S&P 500 warning signal at the 14-day window.

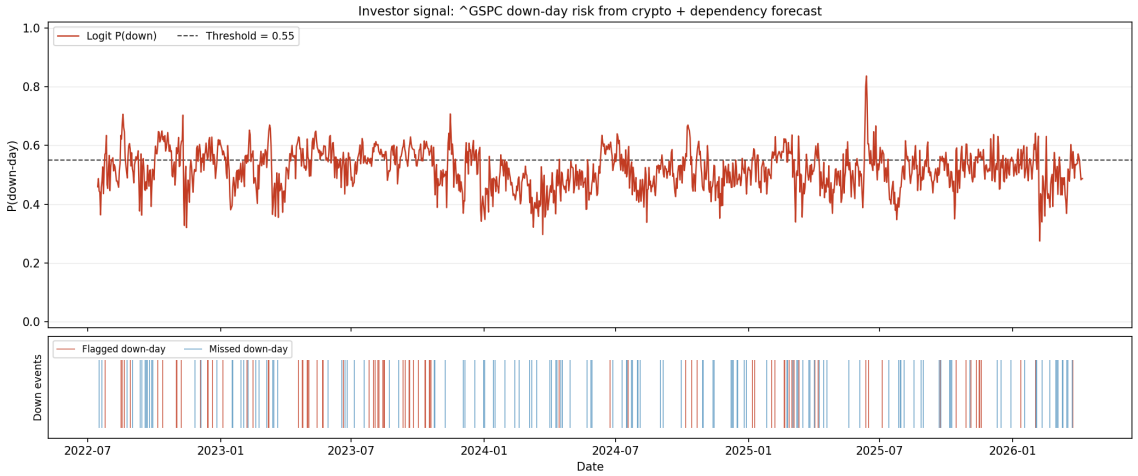


Figure 30: Supplementary S&P 500 warning signal at the 60-day window.

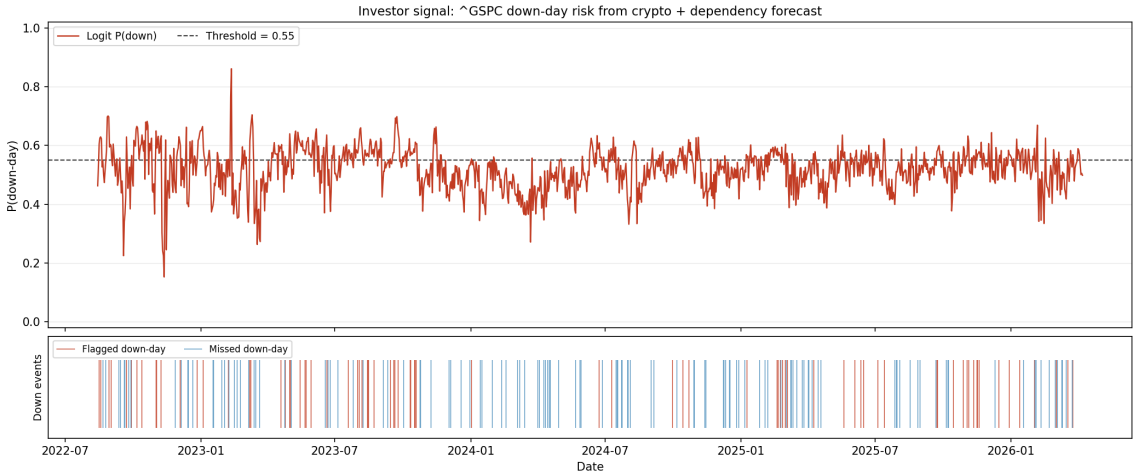


Figure 31: Supplementary S&P 500 warning signal at the 90-day window.

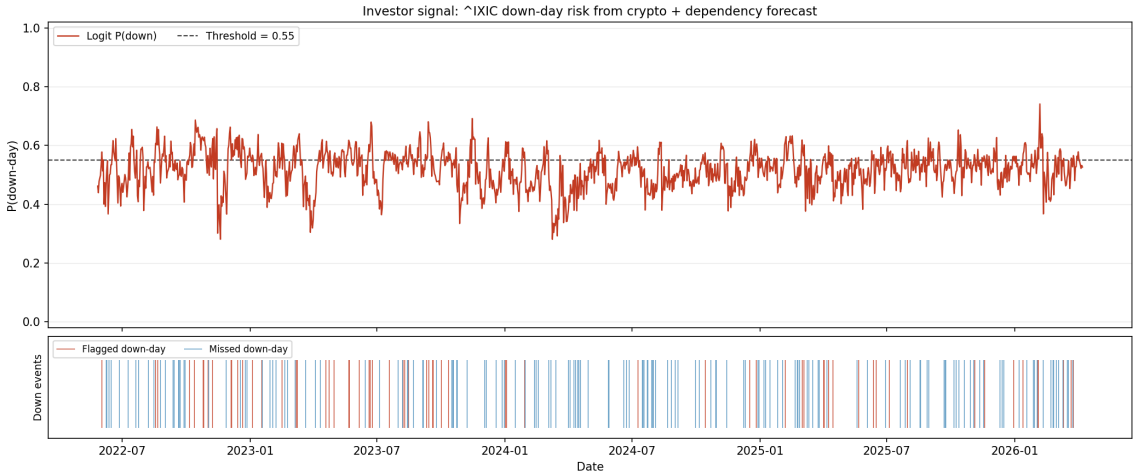


Figure 32: Supplementary NASDAQ warning signal at the 14-day window.

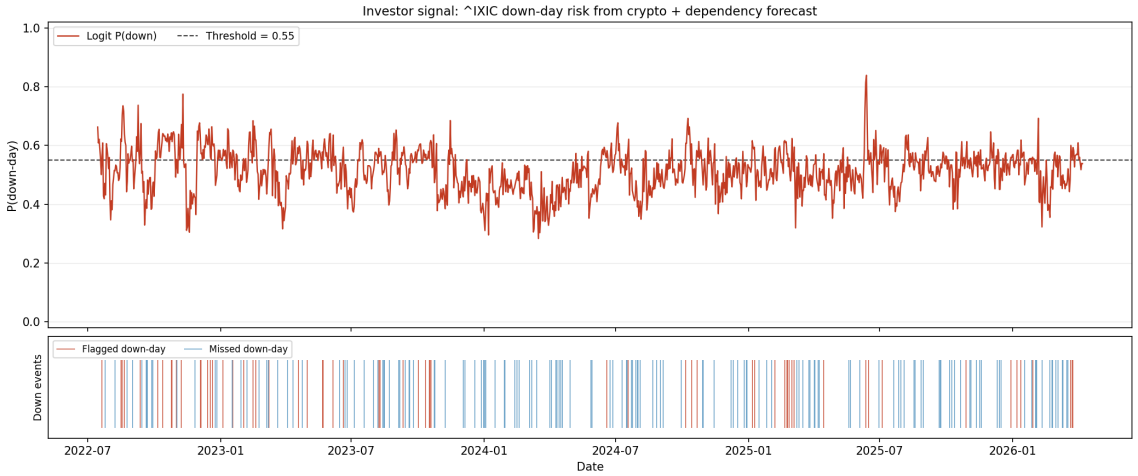


Figure 33: Supplementary NASDAQ warning signal at the 60-day window.

D Appendix D: Reproducibility and Code Structure

D.1 Project organization

The computational part of the thesis is structured as a reproducible research project rather than a collection of disconnected scripts. The main directories are organized as follows:

Listing 5: Top-level project directory structure.

```
clode9/
|-- main.py                 # main pipeline entry point
|-- run_all.py              # full automation: pipeline + notebooks + LaTeX
|-- config.yaml             # experiment configuration
|-- requirements.txt        # Python dependencies
|-- thesis_app/             # core Python modules
|   |-- pipeline.py         # data, features, walk-forward, metrics
|   |-- dcc_walk.py         # leakage-safe DCC-GARCH benchmark
|   |-- signal_layer.py     # investor stress-warning layer
|   |-- notebook_helpers.py # shared plotting and styling
|   |-- data_quality.py     # missing-value and outlier diagnostics
|   `-- regime_analysis.py  # historical regime catalogue
|-- notebooks/              # seven Jupyter notebooks (01-07)
|-- data/raw/               # cached downloaded price data
|-- data/processed/         # derived returns
|-- outputs/results/        # CSV summaries of experiments
|-- outputs/tables/         # LaTeX-exported tables
|-- outputs/figures/        # all generated figures
`-- thesis/                 # this document
```

D.2 Main implementation modules

The following modules constitute the core of the forecasting system:

- `pipeline.py`: orchestrates the full dependency-forecasting and signal-generation workflow.
- `dcc_walk.py`: contains the leakage-safe walk-forward DCC-GARCH implementation.
- `signal_layer.py`: defines the investor warning layer and associated classification metrics.
- `notebook_helpers.py`: shared plotting helpers, consistent styling, and analysis utilities used across all notebooks.
- `data_quality.py`: performs missing-value, outlier, and coverage diagnostics on the raw dataset.
- `regime_analysis.py`: implements the historical regime catalogue and regime-conditional statistics.

This separation of concerns is useful both scientifically and practically. Scientifically, it makes the logic of the workflow easier to audit. Practically, it allows individual modules

to be tested or improved without rewriting the entire system.

D.3 Execution flow

The full pipeline is launched through a single entry point that coordinates data acquisition, model training, notebook execution, and document compilation:

Listing 6: Automated full-pipeline runner (`run_all.py`).

```
import subprocess, sys
from pathlib import Path

BASE = Path(__file__).parent

def run_step(cmd, label):
    print(f"\n=== {label} ===")
    subprocess.run([sys.executable] + cmd, cwd=BASE, check=True)

# Step 1: main pipeline (data download, models, tables, figures)
run_step(["main.py"], "Pipeline")

# Step 2: execute all notebooks in alphabetical order
notebooks = sorted((BASE / "notebooks").glob("*.ipynb"))
for nb in notebooks:
    print(f"  Executing {nb.name}")
    subprocess.run(
        ["jupyter", "nbconvert", "--to", "notebook",
         "--execute", str(nb), "--inplace"],
        cwd=BASE, check=True
    )

# Step 3: compile the thesis PDF
subprocess.run(
    ["latexmk", "-pdf", "-interaction=nonstopmode", "main.tex"],
    cwd=BASE / "thesis", check=True
)
```

At a high level, the execution flow can be summarized as follows:

1. Load configuration and path settings.
2. Load or refresh raw price data from Yahoo Finance.
3. Compute synchronized log returns.
4. Generate rolling-correlation targets for each asset pair and window.
5. Build lagged and volatility-based feature matrices.
6. Fit all models in walk-forward mode and save predictions.
7. Compute metrics, DM tests, and export tables and figures.
8. Run the investor signal layer and export signal figures.

9. Execute six Jupyter notebooks to produce supplementary diagnostics.
10. Compile the LaTeX thesis document.

The main advantage of this design is that every figure and summary reported in the thesis can be traced back to a consistent computational procedure.

D.4 Configuration

Experiment parameters are stored in `config.yaml` and are read once at pipeline startup. This makes it straightforward to change the date range, asset universe, or window set without touching the source code:

Listing 7: Excerpt from `config.yaml` showing key experiment parameters.

```yaml
start_date: "2016-01-01"   # effective start: 2017-11-09 (ETH availability)
end_date:   null           # null = download up to today

assets:
  crypto:      ["BTC-USD", "ETH-USD"]
  traditional: ["^GSPC", "^IXIC", "GLD", "SLV", "UUP"]

base_asset:          "BTC-USD"
rolling_windows:     [14, 30, 60, 90]
forecast_horizon:    1
use_fisher_transform: true
min_train_size:      800
refit_every:         20
xgb_device:          "cuda"
random_state:        42
```

D.5 Verification performed during the project cleanup

During the refinement of the project, several implementation issues were corrected in order to align the code with thesis-level standards.

- A leakage-safe walk-forward DCC benchmark was introduced to avoid full-sample look-ahead contamination.
- The metrics pipeline was made consistent so that all relevant models, including DCC, are evaluated through the same summary path.
- Feature scaling was added for the regularized linear models.
- The XGBoost block was made robust to GPU failure through CPU fallback.
- The investor signal layer was integrated into the main pipeline and exported to dedicated tables and figures.
- Notebook imports and notebook-specific plotting logic were stabilized for presentation and interpretation.

These improvements matter because a thesis is evaluated not only by its conceptual idea but also by whether its empirical implementation is reliable, reproducible, and internally

coherent.

D.6 How to rebuild the thesis artifacts

A practical benefit of the current project state is that the full workflow can be rerun from scratch with a single command from the project root:

Listing 8: Full rebuild from the project root directory.

```
python run_all.py
```

Alternatively, the three stages can be executed individually:

1. Run the main pipeline:
   `python main.py`
   This generates prediction CSVs, metrics, DM-test results, and the majority of figures.
2. Execute the notebooks in order:
   (a) `01_EDA_Dataset.ipynb` — dataset overview and EDA figures.
   (b) `02_GridSearch.ipynb` — cross-validated hyperparameter search.
   (c) `03_Model_Comparison.ipynb` — model-ranking table and heatmaps.
   (d) `04_DM_Tests_Visuals.ipynb` — DM-test heatmap and key forecast figure.
   (e) `05_XGB_vs_DCC.ipynb` — rolling-RMSE, scatter, and error-distribution figures.
   (f) `06_Regime_Analysis.ipynb` — regime catalogue and data-quality diagnostics.
   (g) `07_Robustness_Checks.ipynb` — refit sensitivity, stress threshold sensitivity, bootstrap confidence intervals, signal diagnostics, and cross-asset DM test.
3. Rebuild the PDF document from the `thesis/` directory:

```
pdflatex main.tex && biber main && pdflatex main.tex && pdflatex main.
    ↪ tex
```

Steps 1 and 2 must be completed before step 3 because several figures used in the thesis are produced by the notebooks. This makes the written work maintainable even if the underlying market data are refreshed.

D.7 Final reproducibility note

The value of reproducibility in this thesis is not merely technical. Because the subject concerns financial forecasting, where accidental leakage and selective reporting are constant risks, a transparent and rerunnable codebase directly strengthens the credibility of the empirical conclusions.

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